

# Ch1. Introduction to C++ Programming

```
#include <iostream>
```

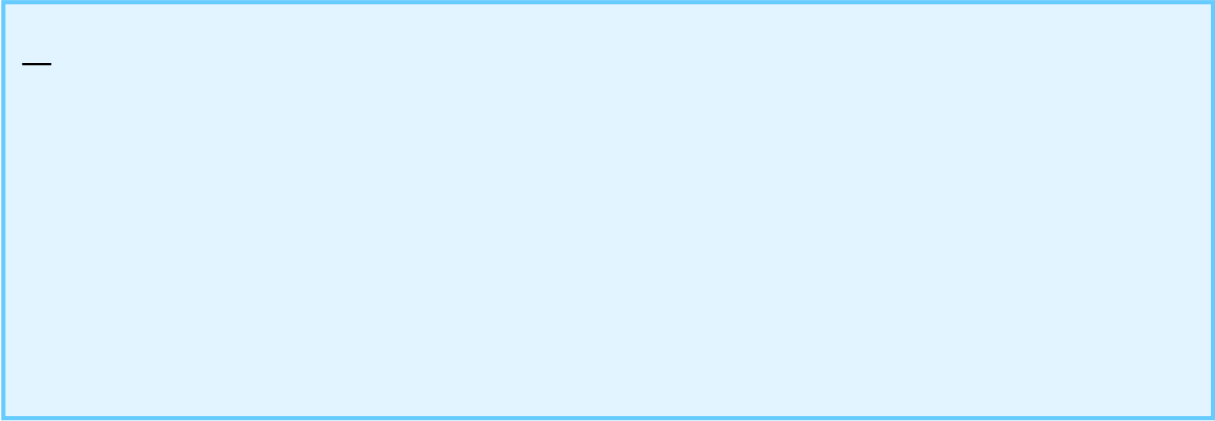
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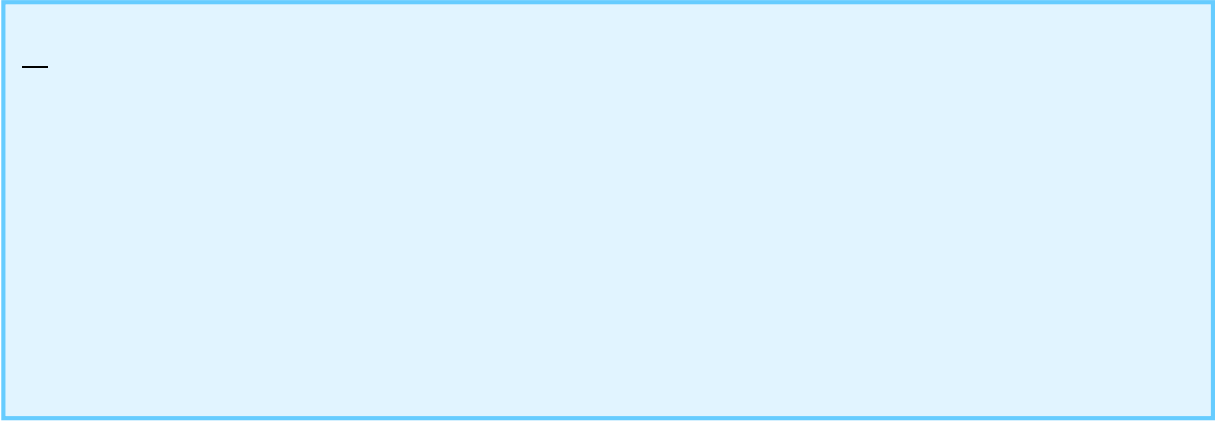
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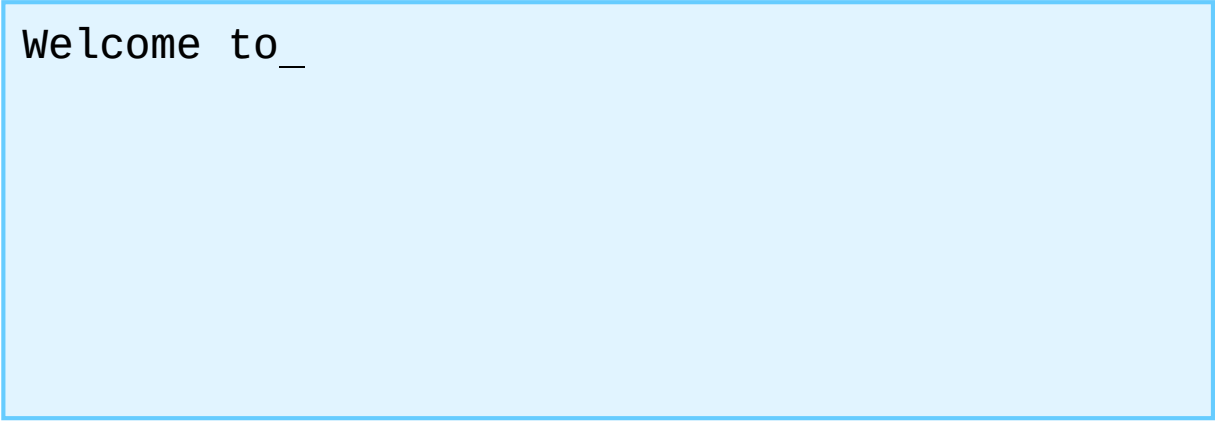
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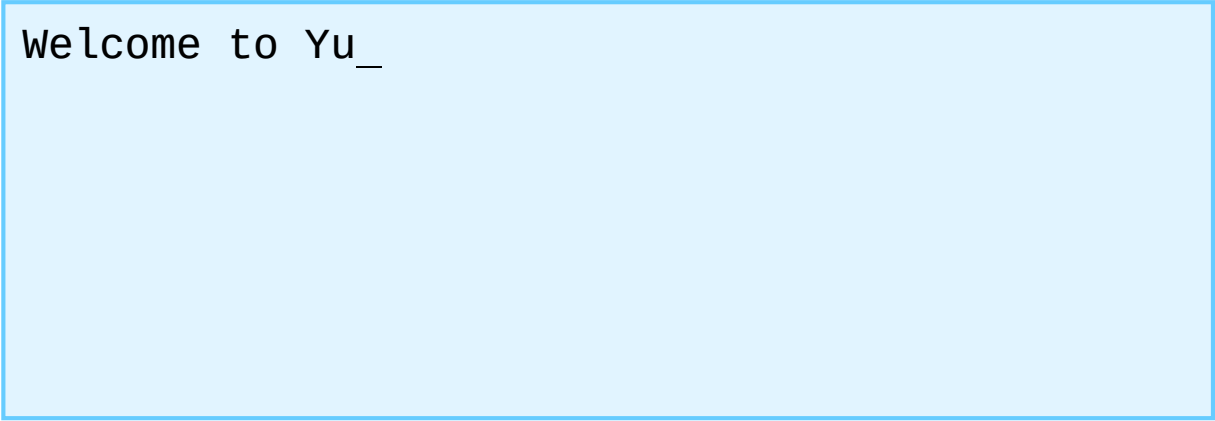
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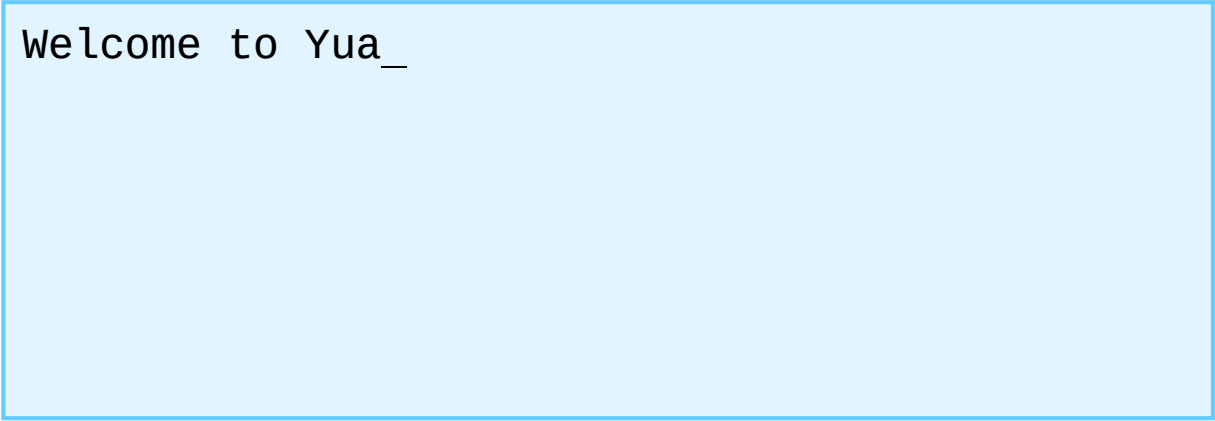
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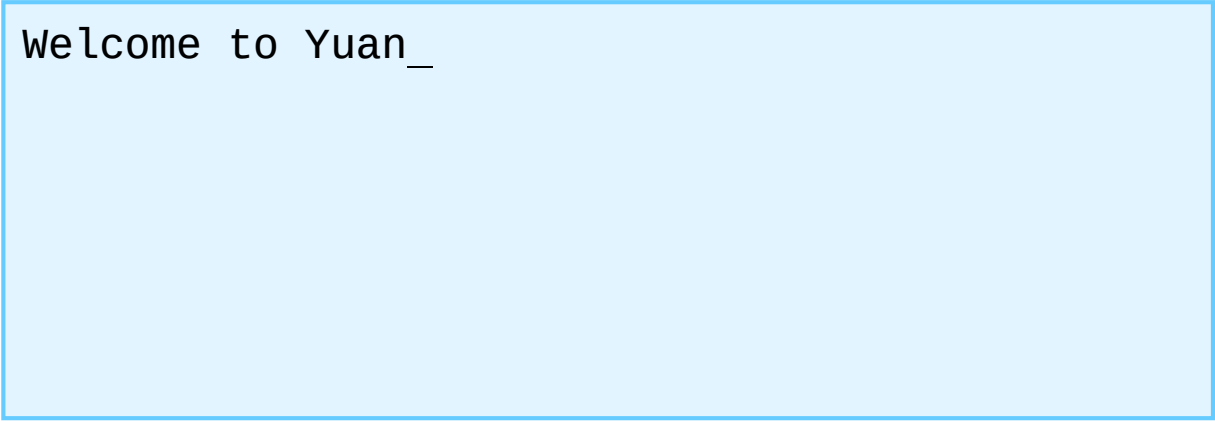
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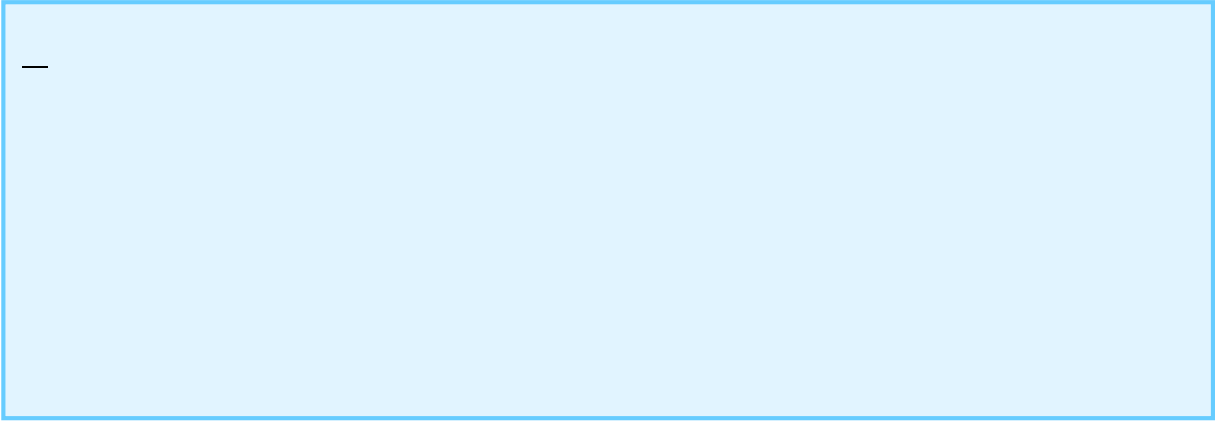
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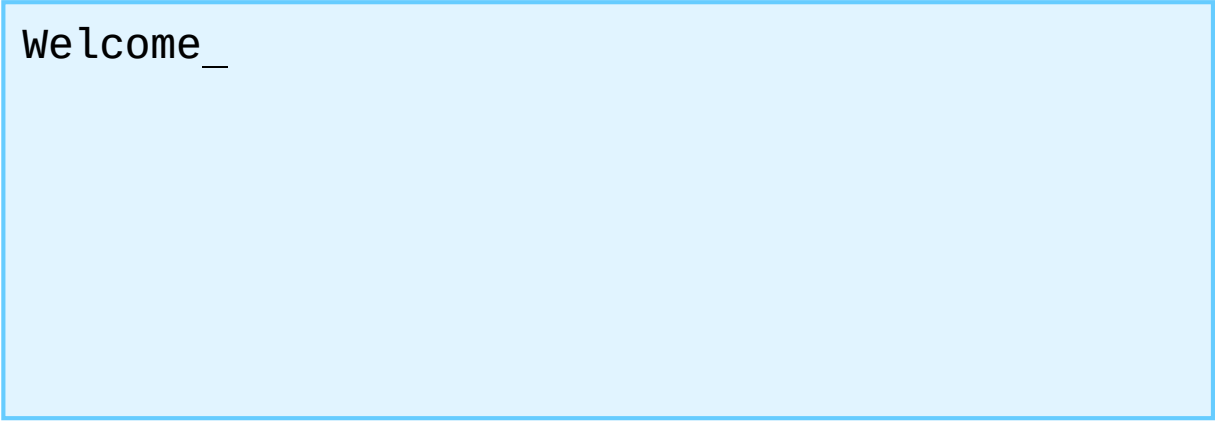
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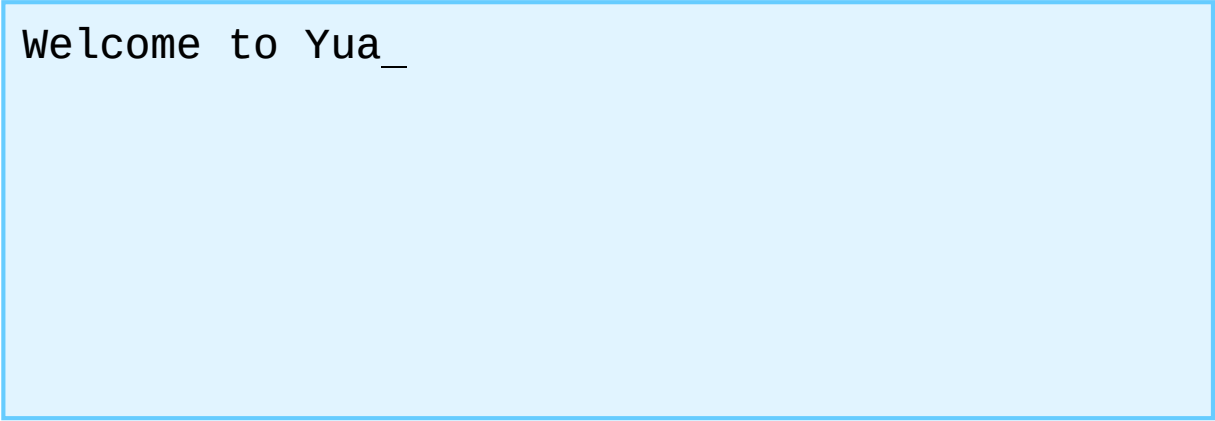
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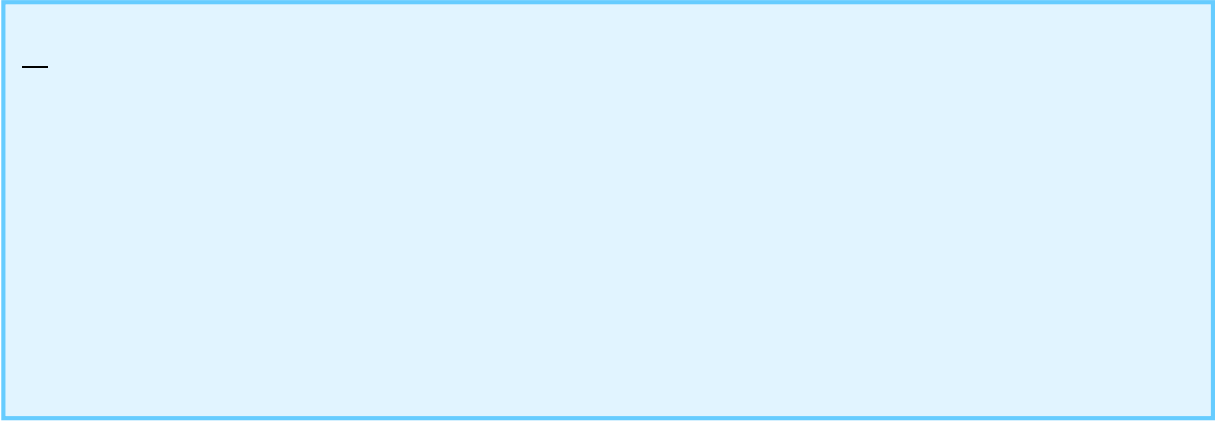
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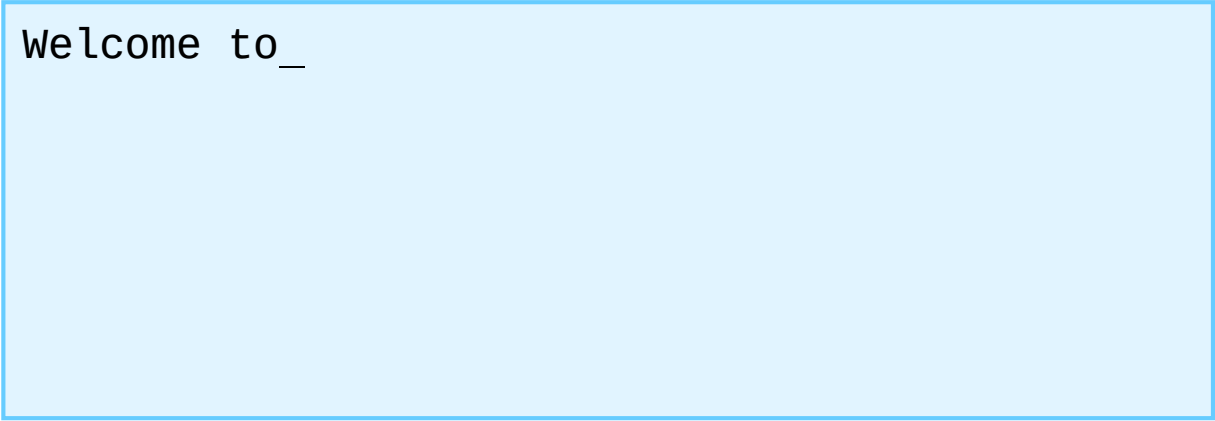
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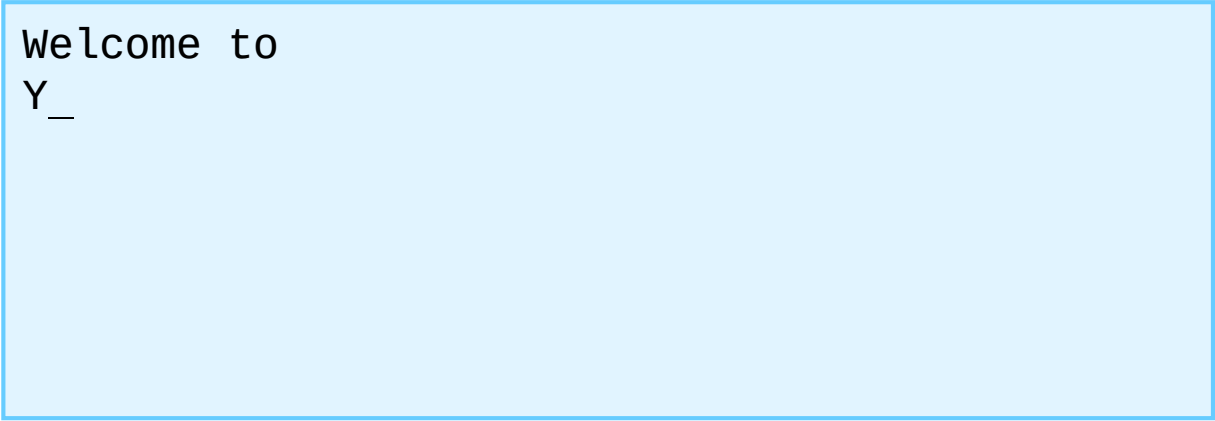
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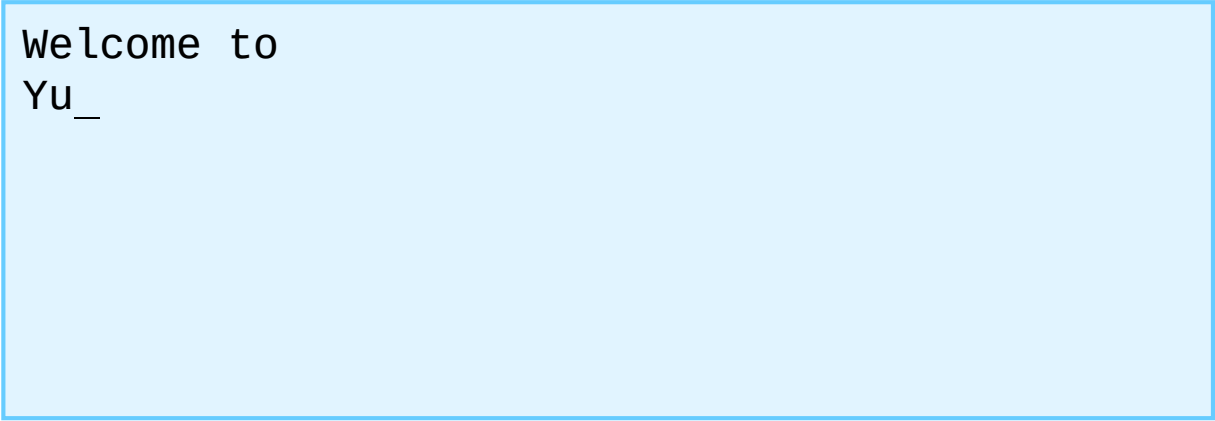
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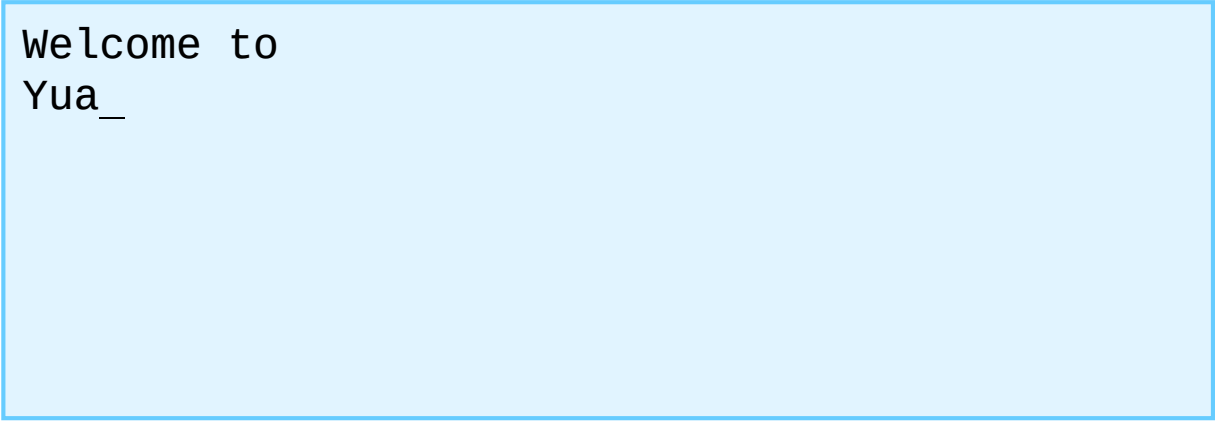
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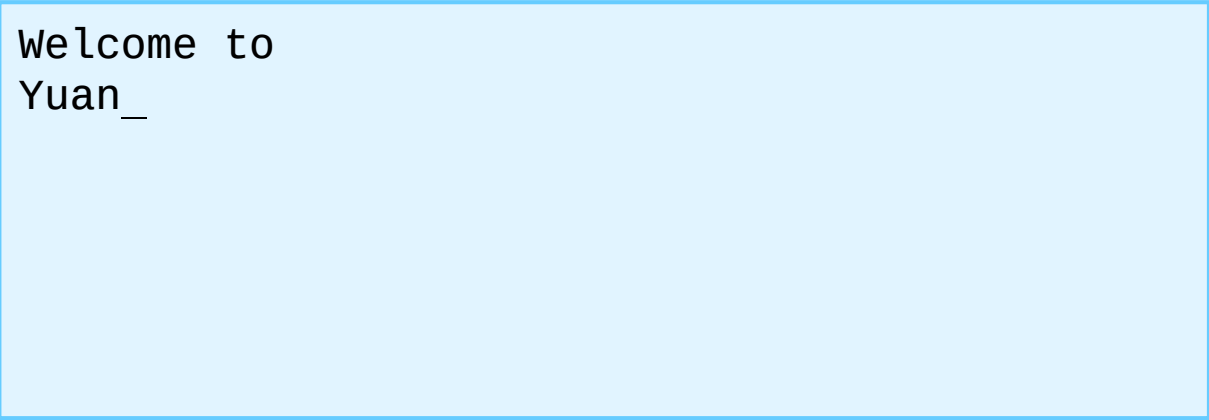
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# Escape sequences

| Escape sequence | Description     |                                                       |
|-----------------|-----------------|-------------------------------------------------------|
| \'              | single quote    | Represents a literal single quotation mark            |
| \"              | double quote    | Represents a literal double quotation mark            |
| \\              | backslash       | Represents a literal backslash character              |
| \a              | audible bell    | Produces an audible alert or beep sound               |
| \b              | backspace       | Moves the cursor back one position                    |
| \n              | new line        | Moves the cursor to the beginning of the next line    |
| \r              | carriage return | Moves the cursor to the beginning of the current line |
| \t              | horizontal tab  | Inserts a horizontal tab space                        |

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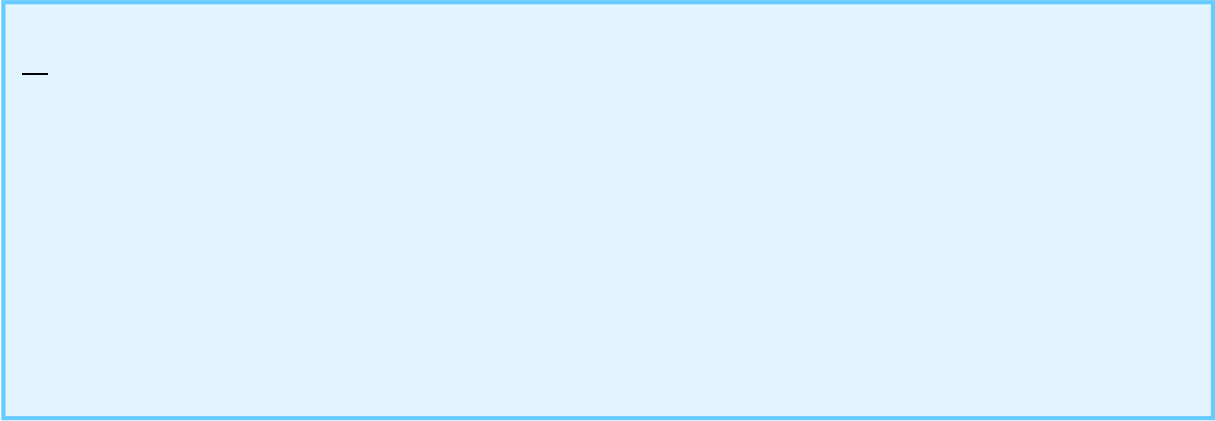
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```
    std::cout << "Yuan\bZe\n";
```

```
    return 0;
```

```
}
```

YuaZe

—



```
#include <iostream>
```

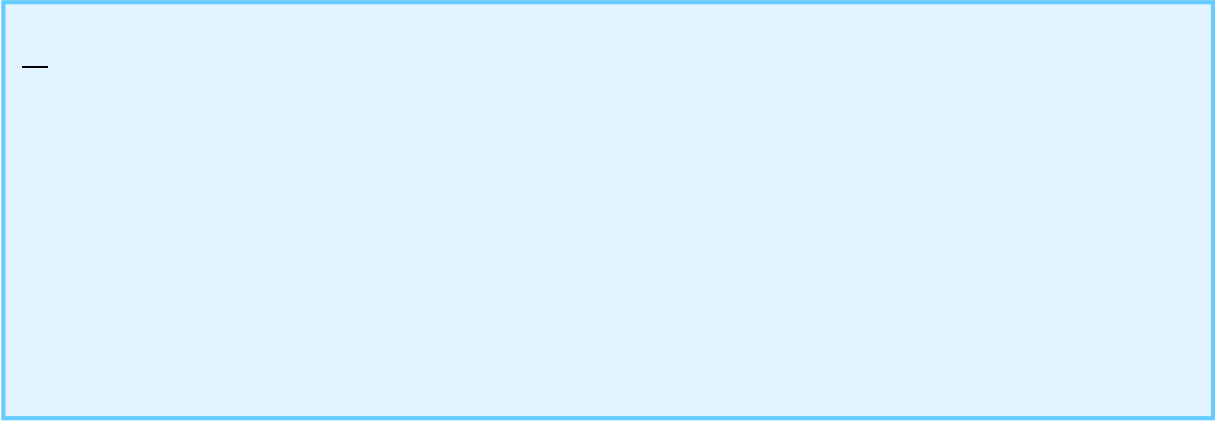
```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```



—



```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Y\_

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Yu\_

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Yua\_

```
#include <iostream>
```

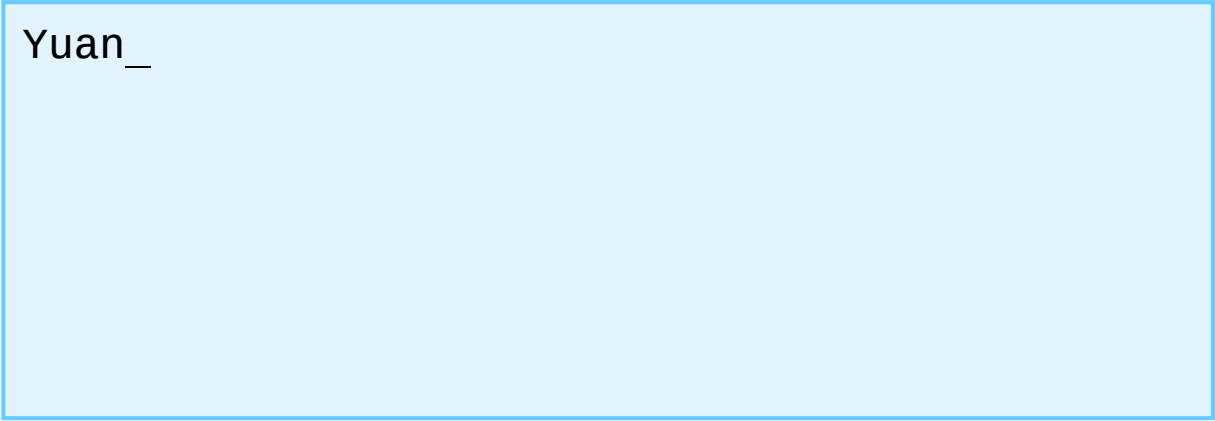
```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```



Yuan\_

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Yuan

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Zuan

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Zean

```
#include <iostream>
```

```
int main()
```

```
{
```

```
    std::cout << "Yuan\rZe\n";
```

```
    return 0;
```

```
}
```

Zean

—



```
#include <iostream>
```

```
int main()
```

```
{
```

```
    // variable declarations
```

```
    int integer1; // declare a variable named integer1
```

```
    int integer2; // declare a variable named integer2
```

```
    int sum; // declare a variable named sum
```

```
    std::cout << "Enter first integer: ";
```

```
    std::cin >> integer1;
```

```
    std::cout << "Enter second integer: ";
```

```
    std::cin >> integer2;
```

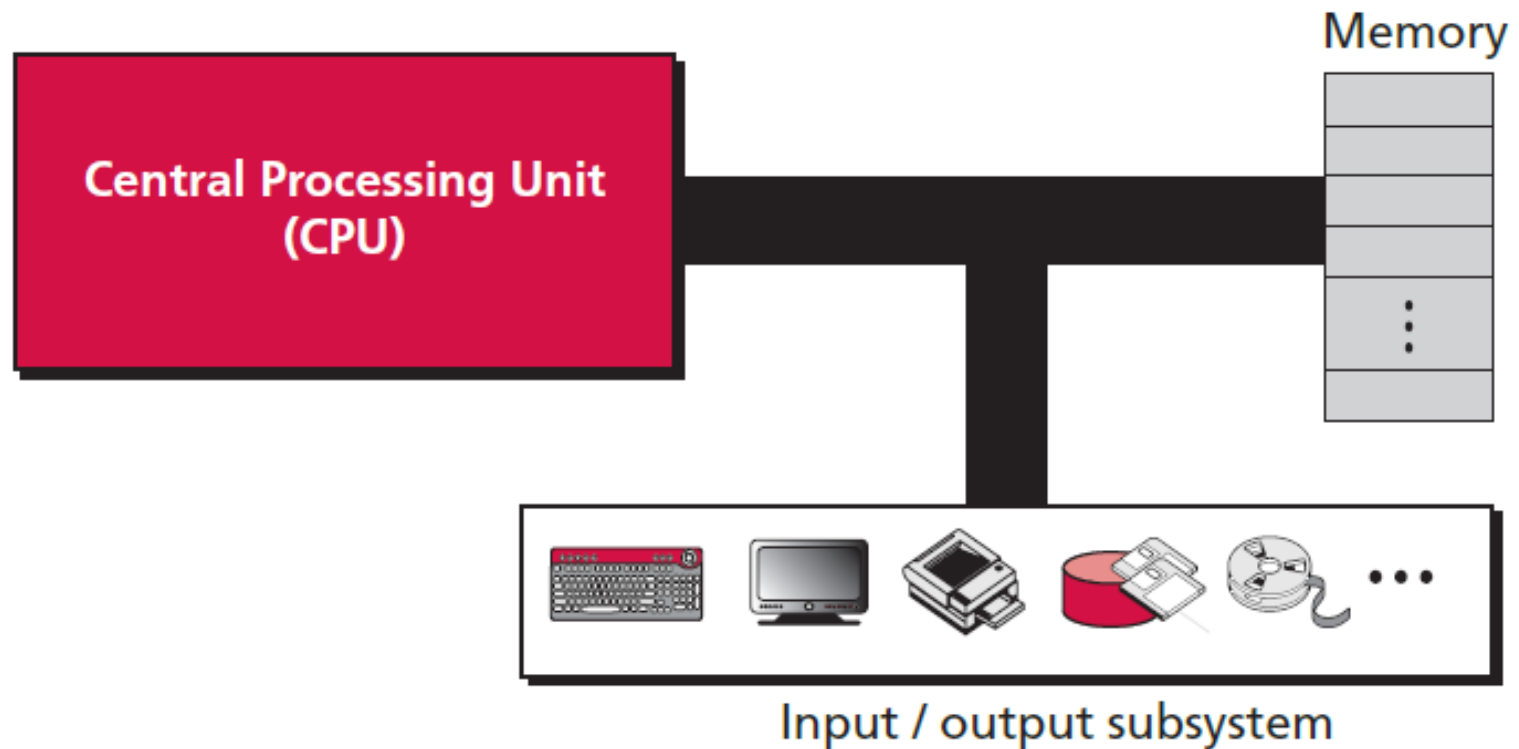
```
    sum = integer1 + integer2;
```

```
    std::cout << "Sum is " << sum << std::endl;
```

```
}
```



# Computer Hardware



values      address

|   |             |
|---|-------------|
|   | 0           |
|   | 1           |
|   | 2           |
|   | 3           |
|   | 4           |
| ⋮ |             |
|   |             |
|   |             |
|   |             |
|   | 17179869181 |
|   | 17179869182 |
|   | 17179869183 |

Main Memory  
(16GB)

values      address

|   |           |
|---|-----------|
|   | 00000000  |
|   | 00000001  |
|   | 00000002  |
|   | 00000003  |
| ⋮ |           |
|   |           |
|   |           |
|   |           |
|   | 00000009  |
|   | 0000000A  |
|   | 0000000B  |
| ⋮ |           |
|   |           |
|   |           |
|   |           |
|   | 3FFFFFFFD |
|   | 3FFFFFFFE |
|   | 3FFFFFFF  |

| binary representation               | hexadecimal | decimal |
|-------------------------------------|-------------|---------|
| 00000000 00000000 00000000 00000000 | 00000000    | 0       |
| 00000000 00000000 00000000 00000001 | 00000001    | 1       |
| 00000000 00000000 00000000 00000010 | 00000002    | 2       |
| 00000000 00000000 00000000 00000011 | 00000003    | 3       |
| 00000000 00000000 00000000 00000100 | 00000004    | 4       |
| 00000000 00000000 00000000 00000101 | 00000005    | 5       |
| 00000000 00000000 00000000 00000110 | 00000006    | 6       |
| 00000000 00000000 00000000 00000111 | 00000007    | 7       |
| 00000000 00000000 00000000 00001000 | 00000008    | 8       |
| 00000000 00000000 00000000 00001001 | 00000009    | 9       |
| 00000000 00000000 00000000 00001010 | 0000000A    | 10      |
| 00000000 00000000 00000000 00001011 | 0000000B    | 11      |
| 00000000 00000000 00000000 00001100 | 0000000C    | 12      |
| 00000000 00000000 00000000 00001101 | 0000000D    | 13      |
| 00000000 00000000 00000000 00001110 | 0000000E    | 14      |
| 00000000 00000000 00000000 00001111 | 0000000F    | 15      |

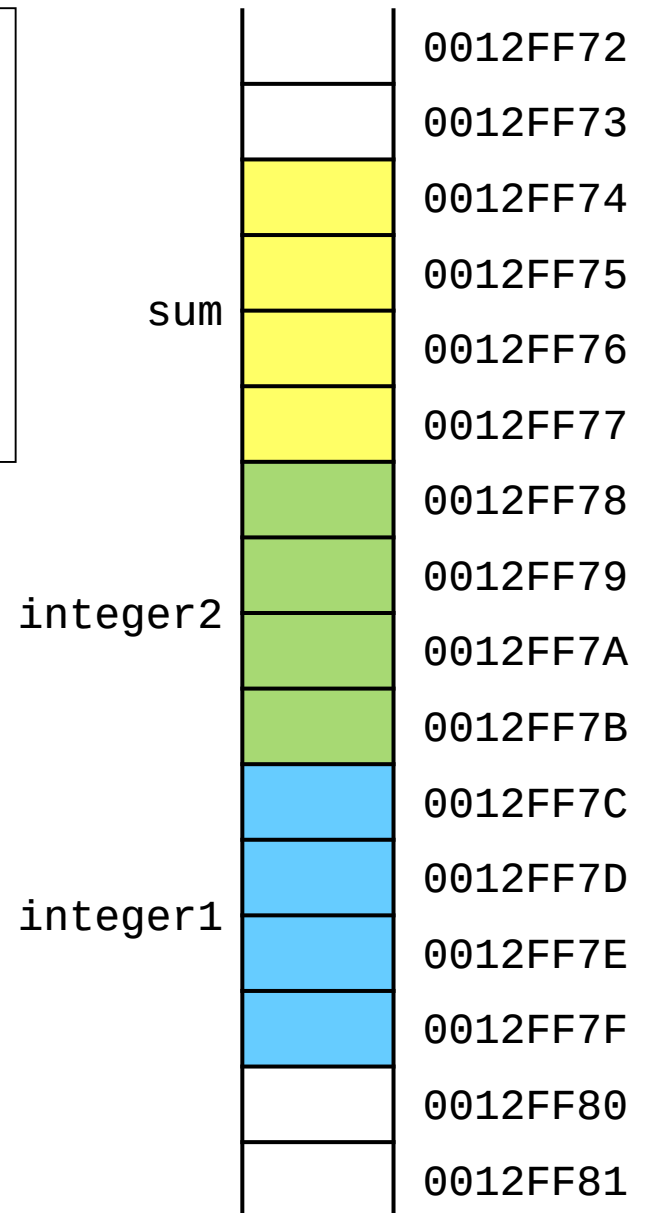
| binary representation               | hexadecimal | decimal |
|-------------------------------------|-------------|---------|
| 00000000 00000000 00000000 00010000 | 00000010    | 16      |
| 00000000 00000000 00000000 00010001 | 00000011    | 17      |
| 00000000 00000000 00000000 00010010 | 00000012    | 18      |
| 00000000 00000000 00000000 00010011 | 00000013    | 19      |
| 00000000 00000000 00000000 00010100 | 00000014    | 20      |
| 00000000 00000000 00000000 00010101 | 00000015    | 21      |
| 00000000 00000000 00000000 00010110 | 00000016    | 22      |
| 00000000 00000000 00000000 00010111 | 00000017    | 23      |
| 00000000 00000000 00000000 00011000 | 00000018    | 24      |
| 00000000 00000000 00000000 00011001 | 00000019    | 25      |
| 00000000 00000000 00000000 00011010 | 0000001A    | 26      |
| 00000000 00000000 00000000 00011011 | 0000001B    | 27      |
| 00000000 00000000 00000000 00011100 | 0000001C    | 28      |
| 00000000 00000000 00000000 00011101 | 0000001D    | 29      |
| 00000000 00000000 00000000 00011110 | 0000001E    | 30      |
| 00000000 00000000 00000000 00011111 | 0000001F    | 31      |

| binary representation               | hexadecimal | decimal    |
|-------------------------------------|-------------|------------|
| 00000000 00000000 00000000 00100000 | 00000020    | 32         |
| 00000000 00000000 00000000 00100001 | 00000021    | 33         |
| 00000000 00000000 00000000 00100010 | 00000022    | 34         |
| 00000000 00000000 00000000 00100011 | 00000023    | 35         |
| . . . . .                           | . . . . .   | . . . . .  |
| 00000000 00010010 11111111 01110100 | 0012FF74    | 1245044    |
| 00000000 00010010 11111111 01111000 | 0012FF78    | 1245048    |
| 00000000 00010010 11111111 01111100 | 0012FF7C    | 1245052    |
| 00000000 00010010 11111111 10000000 | 0012FF80    | 1245056    |
| . . . . .                           | . . . . .   | . . . . .  |
| 11111111 11111111 11111111 11111010 | FFFFFFFA    | 4294967290 |
| 11111111 11111111 11111111 11111011 | FFFFFFFB    | 4294967291 |
| 11111111 11111111 11111111 11111100 | FFFFFFFC    | 4294967292 |
| 11111111 11111111 11111111 11111101 | FFFFFFFD    | 4294967293 |
| 11111111 11111111 11111111 11111110 | FFFFFFFE    | 4294967294 |
| 11111111 11111111 11111111 11111111 | FFFFFFF     | 4294967295 |

```
int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;
```



Output





```
int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;
```



Output

|          |  |          |
|----------|--|----------|
| sum      |  | 0012FF72 |
|          |  | 0012FF73 |
|          |  | 0012FF74 |
|          |  | 0012FF75 |
|          |  | 0012FF76 |
| integer2 |  | 0012FF77 |
|          |  | 0012FF78 |
|          |  | 0012FF79 |
|          |  | 0012FF7A |
| integer1 |  | 0012FF7B |
|          |  | 0012FF7C |
|          |  | 0012FF7D |
|          |  | 0012FF7E |
|          |  | 0012FF7F |
|          |  | 0012FF80 |
|          |  | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

Enter first integer: \_

Output

|          |  |          |
|----------|--|----------|
| sum      |  | 0012FF72 |
|          |  | 0012FF73 |
|          |  | 0012FF74 |
|          |  | 0012FF75 |
|          |  | 0012FF76 |
| integer2 |  | 0012FF77 |
|          |  | 0012FF78 |
|          |  | 0012FF79 |
|          |  | 0012FF7A |
| integer1 |  | 0012FF7B |
|          |  | 0012FF7C |
|          |  | 0012FF7D |
|          |  | 0012FF7E |
|          |  | 0012FF7F |
|          |  | 0012FF80 |
|          |  | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

Enter first integer: 28\_

Output

|          |  |          |
|----------|--|----------|
| sum      |  | 0012FF72 |
|          |  | 0012FF73 |
|          |  | 0012FF74 |
|          |  | 0012FF75 |
|          |  | 0012FF76 |
| integer2 |  | 0012FF77 |
|          |  | 0012FF78 |
|          |  | 0012FF79 |
|          |  | 0012FF7A |
| integer1 |  | 0012FF7B |
|          |  | 0012FF7C |
|          |  | 0012FF7D |
|          |  | 0012FF7E |
|          |  | 0012FF7F |
|          |  | 0012FF80 |
|          |  | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28

```

```

—

```

Output

sum

integer2

integer1

|  |          |
|--|----------|
|  | 0012FF72 |
|  | 0012FF73 |
|  | 0012FF74 |
|  | 0012FF75 |
|  | 0012FF76 |
|  | 0012FF77 |
|  | 0012FF78 |
|  | 0012FF79 |
|  | 0012FF7A |
|  | 0012FF7B |
|  | 0012FF7C |
|  | 0012FF7D |
|  | 0012FF7E |
|  | 0012FF7F |
|  | 0012FF80 |
|  | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

Enter first integer: 28

—

Output

|          |    |          |
|----------|----|----------|
| sum      |    | 0012FF72 |
|          |    | 0012FF73 |
|          |    | 0012FF74 |
|          |    | 0012FF75 |
|          |    | 0012FF76 |
| integer2 |    | 0012FF77 |
|          |    | 0012FF78 |
|          |    | 0012FF79 |
|          |    | 0012FF7A |
| integer1 |    | 0012FF7B |
|          |    | 0012FF7C |
|          | 28 | 0012FF7D |
|          |    | 0012FF7E |
|          |    | 0012FF7F |
|          |    | 0012FF80 |
|          |    | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: _

```

Output

|          |    |          |
|----------|----|----------|
| sum      |    | 0012FF72 |
|          |    | 0012FF73 |
|          |    | 0012FF74 |
|          |    | 0012FF75 |
|          |    | 0012FF76 |
| integer2 |    | 0012FF77 |
|          |    | 0012FF78 |
|          |    | 0012FF79 |
|          |    | 0012FF7A |
| integer1 |    | 0012FF7B |
|          |    | 0012FF7C |
|          | 28 | 0012FF7D |
|          |    | 0012FF7E |
|          |    | 0012FF7F |
|          |    | 0012FF80 |
|          |    | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96_

```

Output

|          |    |          |
|----------|----|----------|
| sum      |    | 0012FF72 |
|          |    | 0012FF73 |
|          |    | 0012FF74 |
|          |    | 0012FF75 |
|          |    | 0012FF76 |
| integer2 |    | 0012FF77 |
|          |    | 0012FF78 |
|          |    | 0012FF79 |
|          |    | 0012FF7A |
| integer1 |    | 0012FF7B |
|          |    | 0012FF7C |
|          | 28 | 0012FF7D |
|          |    | 0012FF7E |
|          |    | 0012FF7F |
|          |    | 0012FF80 |
|          |    | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
—

```

Output

|          |    |          |
|----------|----|----------|
| sum      |    | 0012FF72 |
|          |    | 0012FF73 |
|          |    | 0012FF74 |
|          |    | 0012FF75 |
| integer2 |    | 0012FF76 |
|          |    | 0012FF77 |
|          |    | 0012FF78 |
|          |    | 0012FF79 |
| integer1 |    | 0012FF7A |
|          |    | 0012FF7B |
|          | 28 | 0012FF7C |
|          |    | 0012FF7D |
|          |    | 0012FF7E |
|          |    | 0012FF7F |
|          |    | 0012FF80 |
|          |    | 0012FF81 |



```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
—

```

Output

|          |    |          |
|----------|----|----------|
| sum      |    | 0012FF72 |
|          |    | 0012FF73 |
|          |    | 0012FF74 |
|          |    | 0012FF75 |
|          |    | 0012FF76 |
| integer2 | 96 | 0012FF77 |
|          |    | 0012FF78 |
|          |    | 0012FF79 |
|          |    | 0012FF7A |
| integer1 | 28 | 0012FF7B |
|          |    | 0012FF7C |
|          |    | 0012FF7D |
|          |    | 0012FF7E |
|          |    | 0012FF7F |
|          |    | 0012FF80 |
|          |    | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
—

```

Output

|          |     |          |
|----------|-----|----------|
| sum      | 124 | 0012FF72 |
|          |     | 0012FF73 |
|          |     | 0012FF74 |
|          |     | 0012FF75 |
| integer2 | 96  | 0012FF76 |
|          |     | 0012FF77 |
|          |     | 0012FF78 |
|          |     | 0012FF79 |
| integer1 | 28  | 0012FF7A |
|          |     | 0012FF7B |
|          |     | 0012FF7C |
|          |     | 0012FF7D |
|          |     | 0012FF7E |
|          |     | 0012FF7F |
|          |     | 0012FF80 |
|          |     | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
Sum is _

```

Output

|          |     |          |
|----------|-----|----------|
| sum      | 124 | 0012FF72 |
|          |     | 0012FF73 |
|          |     | 0012FF74 |
|          |     | 0012FF75 |
|          |     | 0012FF76 |
| integer2 | 96  | 0012FF77 |
|          |     | 0012FF78 |
|          |     | 0012FF79 |
|          |     | 0012FF7A |
| integer1 | 28  | 0012FF7B |
|          |     | 0012FF7C |
|          |     | 0012FF7D |
|          |     | 0012FF7E |
|          |     | 0012FF7F |
|          |     | 0012FF80 |
|          |     | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
Sum is 124_

```

Output

|          |     |          |
|----------|-----|----------|
| sum      | 124 | 0012FF72 |
|          |     | 0012FF73 |
|          |     | 0012FF74 |
|          |     | 0012FF75 |
| integer2 | 96  | 0012FF76 |
|          |     | 0012FF77 |
|          |     | 0012FF78 |
|          |     | 0012FF79 |
| integer1 | 28  | 0012FF7A |
|          |     | 0012FF7B |
|          |     | 0012FF7C |
|          |     | 0012FF7D |
|          |     | 0012FF7E |
|          |     | 0012FF7F |
|          |     | 0012FF80 |
|          |     | 0012FF81 |

```

int integer1;  int integer2;  int sum;
std::cout << "Enter first integer: ";
std::cin >> integer1;
std::cout << "Enter second integer: ";
std::cin >> integer2;
sum = integer1 + integer2;
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
Sum is 124

```

Output

|          |     |          |
|----------|-----|----------|
| sum      | 124 | 0012FF72 |
|          |     | 0012FF73 |
|          |     | 0012FF74 |
|          |     | 0012FF75 |
| integer2 | 96  | 0012FF76 |
|          |     | 0012FF77 |
|          |     | 0012FF78 |
|          |     | 0012FF79 |
| integer1 | 28  | 0012FF7A |
|          |     | 0012FF7B |
|          |     | 0012FF7C |
|          |     | 0012FF7D |
|          |     | 0012FF7E |
|          |     | 0012FF7F |
|          |     | 0012FF80 |
|          |     | 0012FF81 |

```
#include <iostream>
using namespace std;

int main()
{
    // array declarations
    int integers[ 2 ]; // an array
    int sum;

    cout << "Enter first integer: ";
    cin >> integers[ 0 ];

    cout << "Enter second integer: ";
    cin >> integers[ 1 ];

    sum = integers[ 0 ] + integers[ 1 ];

    cout << "Sum is " << sum << endl;
}
```

```

int integers[ 2 ];  int sum;
std::cout << "Enter first integer: ";
std::cin >> integers[ 0 ];
std::cout << "Enter second integer: ";
std::cin >> integers[ 1 ];
sum = integers[ 0 ] + integers[ 1 ];
std::cout << "Sum is " << sum << std::endl;

```

Enter first integer: 28\_

Output

integers[0]

integers[1]

sum

|  |          |
|--|----------|
|  | 0012FF72 |
|  | 0012FF73 |
|  | 0012FF74 |
|  | 0012FF75 |
|  | 0012FF76 |
|  | 0012FF77 |
|  | 0012FF78 |
|  | 0012FF79 |
|  | 0012FF7A |
|  | 0012FF7B |
|  | 0012FF7C |
|  | 0012FF7D |
|  | 0012FF7E |
|  | 0012FF7F |
|  | 0012FF80 |
|  | 0012FF81 |

```

int integers[ 2 ];  int sum;
std::cout << "Enter first integer: ";
std::cin >> integers[ 0 ];
std::cout << "Enter second integer: ";
std::cin >> integers[ 1 ];
sum = integers[ 0 ] + integers[ 1 ];
std::cout << "Sum is " << sum << std::endl;

```

Enter first integer: 28

—

Output

integers[0]

28

integers[1]

sum

|  |          |
|--|----------|
|  | 0012FF72 |
|  | 0012FF73 |
|  | 0012FF74 |
|  | 0012FF75 |
|  | 0012FF76 |
|  | 0012FF77 |
|  | 0012FF78 |
|  | 0012FF79 |
|  | 0012FF7A |
|  | 0012FF7B |
|  | 0012FF7C |
|  | 0012FF7D |
|  | 0012FF7E |
|  | 0012FF7F |
|  | 0012FF80 |
|  | 0012FF81 |



```

int integers[ 2 ];  int sum;
std::cout << "Enter first integer: ";
std::cin >> integers[ 0 ];
std::cout << "Enter second integer: ";
std::cin >> integers[ 1 ];
sum = integers[ 0 ] + integers[ 1 ];
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96_

```

Output

integers[0]

28

integers[1]

sum

|  |          |
|--|----------|
|  | 0012FF72 |
|  | 0012FF73 |
|  | 0012FF74 |
|  | 0012FF75 |
|  | 0012FF76 |
|  | 0012FF77 |
|  | 0012FF78 |
|  | 0012FF79 |
|  | 0012FF7A |
|  | 0012FF7B |
|  | 0012FF7C |
|  | 0012FF7D |
|  | 0012FF7E |
|  | 0012FF7F |
|  | 0012FF80 |
|  | 0012FF81 |

```

int integers[ 2 ];  int sum;
std::cout << "Enter first integer: ";
std::cin >> integers[ 0 ];
std::cout << "Enter second integer: ";
std::cin >> integers[ 1 ];
sum = integers[ 0 ] + integers[ 1 ];
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
—

```

Output

|             |    |          |
|-------------|----|----------|
| sum         |    | 0012FF72 |
|             |    | 0012FF73 |
|             |    | 0012FF74 |
|             |    | 0012FF75 |
|             |    | 0012FF76 |
| integers[0] | 28 | 0012FF77 |
|             |    | 0012FF78 |
|             |    | 0012FF79 |
|             |    | 0012FF7A |
|             |    | 0012FF7B |
| integers[1] | 96 | 0012FF7C |
|             |    | 0012FF7D |
|             |    | 0012FF7E |
|             |    | 0012FF7F |
|             |    | 0012FF80 |
|             |    | 0012FF81 |

```
int integers[ 2 ];  int sum;
std::cout << "Enter first integer: ";
std::cin >> integers[ 0 ];
std::cout << "Enter second integer: ";
std::cin >> integers[ 1 ];
sum = integers[ 0 ] + integers[ 1 ];
std::cout << "Sum is " << sum << std::endl;
```

```
Enter first integer: 28
Enter second integer: 96
—
```

Output

|             |     |          |
|-------------|-----|----------|
| sum         | 124 | 0012FF72 |
|             |     | 0012FF73 |
|             |     | 0012FF74 |
|             |     | 0012FF75 |
| integers[0] | 28  | 0012FF76 |
|             |     | 0012FF77 |
|             |     | 0012FF78 |
|             |     | 0012FF79 |
| integers[1] | 96  | 0012FF7A |
|             |     | 0012FF7B |
|             |     | 0012FF7C |
|             |     | 0012FF7D |
|             |     | 0012FF7E |
|             |     | 0012FF7F |
|             |     | 0012FF80 |
|             |     | 0012FF81 |

```

int integers[ 2 ];  int sum;
std::cout << "Enter first integer: ";
std::cin >> integers[ 0 ];
std::cout << "Enter second integer: ";
std::cin >> integers[ 1 ];
sum = integers[ 0 ] + integers[ 1 ];
std::cout << "Sum is " << sum << std::endl;

```

```

Enter first integer: 28
Enter second integer: 96
Sum is 124

```

Output

|             |     |          |
|-------------|-----|----------|
| sum         | 124 | 0012FF72 |
|             |     | 0012FF73 |
|             |     | 0012FF74 |
|             |     | 0012FF75 |
| integers[0] | 28  | 0012FF76 |
|             |     | 0012FF77 |
|             |     | 0012FF78 |
|             |     | 0012FF79 |
| integers[1] | 96  | 0012FF7A |
|             |     | 0012FF7B |
|             |     | 0012FF7C |
|             |     | 0012FF7D |
|             |     | 0012FF7E |
|             |     | 0012FF7F |
|             |     | 0012FF80 |
|             |     | 0012FF81 |

```
int number[ 8 ]; // eight integers
```

```
int number1; // first integer  
int number2; // second integer  
int number3; // third integer  
int number4; // fourth integer  
int number5; // fifth integer  
int number6; // sixth integer  
int number7; // seventh integer  
int number8; // eighth integer
```

```
number[ 5 ] = 10;
```

```
number3 = 20;
```

|         |  |          |
|---------|--|----------|
| number8 |  | 0012FF40 |
| number7 |  | 0012FF44 |
| number6 |  | 0012FF48 |
| number5 |  | 0012FF4C |
| number4 |  | 0012FF50 |
| number3 |  | 0012FF54 |
| number2 |  | 0012FF58 |
| number1 |  | 0012FF5C |

|           |  |          |
|-----------|--|----------|
| number[0] |  | 0012FF60 |
| number[1] |  | 0012FF64 |
| number[2] |  | 0012FF68 |
| number[3] |  | 0012FF6C |
| number[4] |  | 0012FF70 |
| number[5] |  | 0012FF74 |
| number[6] |  | 0012FF78 |
| number[7] |  | 0012FF7C |

```
int number[ 8 ]; // eight integers
```

```
int number1; // first integer  
int number2; // second integer  
int number3; // third integer  
int number4; // fourth integer  
int number5; // fifth integer  
int number6; // sixth integer  
int number7; // seventh integer  
int number8; // eighth integer
```

```
number[ 5 ] = 10;
```

```
number3 = 20;
```

|         |  |          |
|---------|--|----------|
| number8 |  | 0012FF40 |
| number7 |  | 0012FF44 |
| number6 |  | 0012FF48 |
| number5 |  | 0012FF4C |
| number4 |  | 0012FF50 |
| number3 |  | 0012FF54 |
| number2 |  | 0012FF58 |
| number1 |  | 0012FF5C |

|           |    |          |
|-----------|----|----------|
| number[0] |    | 0012FF60 |
| number[1] |    | 0012FF64 |
| number[2] |    | 0012FF68 |
| number[3] |    | 0012FF6C |
| number[4] |    | 0012FF70 |
| number[5] | 10 | 0012FF74 |
| number[6] |    | 0012FF78 |
| number[7] |    | 0012FF7C |

```
int number[ 8 ]; // eight integers
```

```
int number1; // first integer  
int number2; // second integer  
int number3; // third integer  
int number4; // fourth integer  
int number5; // fifth integer  
int number6; // sixth integer  
int number7; // seventh integer  
int number8; // eighth integer
```

```
number[ 5 ] = 10;
```

```
number3 = 20;
```

|         |    |          |
|---------|----|----------|
| number8 |    | 0012FF40 |
| number7 |    | 0012FF44 |
| number6 |    | 0012FF48 |
| number5 |    | 0012FF4C |
| number4 |    | 0012FF50 |
| number3 | 20 | 0012FF54 |
| number2 |    | 0012FF58 |
| number1 |    | 0012FF5C |

|           |    |          |
|-----------|----|----------|
| number[0] |    | 0012FF60 |
| number[1] |    | 0012FF64 |
| number[2] |    | 0012FF68 |
| number[3] |    | 0012FF6C |
| number[4] |    | 0012FF70 |
| number[5] | 10 | 0012FF74 |
| number[6] |    | 0012FF78 |
| number[7] |    | 0012FF7C |

# Arithmetic Operators

```
int main()
{
    int a, b;
    a = 7;
    b = 2;

    // printing the sum of a and b
    cout << "a + b = " << ( a + b ) << endl;

    // printing the difference of a and b
    cout << "a - b = " << ( a - b ) << endl;

    // printing the product of a and b
    cout << "a * b = " << ( a * b ) << endl;

    // printing the division of a by b
    cout << "a / b = " << ( a / b ) << endl;

    // printing the modulo of a by b
    cout << "a % b = " << ( a % b ) << endl;
}
```



$$a + b = 9$$

$$a - b = 5$$

$$a * b = 14$$

$$a / b = 3$$

$$a \% b = 1$$

# Initialization

```
int main()
{
    int a = 5; // initializer after equals sign
    int b( 6 ); // initializer in parenthesis
    int c{ 7 }; // initializer in braces
    int d = { 8 }; // initializer in braces
    int e{}; // initializer in braces

    cout << "a: " << a << endl;
    cout << "b: " << b << endl;
    cout << "c: " << c << endl;
    cout << "d: " << d << endl;
    cout << "e: " << e << endl;
    cout << "int(): " << int() << endl;
}
```

## Output

```
a: 5
b: 6
c: 7
d: 8
e: 0
int(): 0
```

# Initialization

```
int main()
{
    int a = int();
    int c{ int() };
    int d = { int() };
    int e;

    cout << "a: " << a << endl;
    cout << "c: " << c << endl;
    cout << "d: " << d << endl;
    e = int();
    cout << "e: " << e << endl;
    cout << "int(): " << int() << endl;
}
```

## Output

```
a: 0
c: 0
d: 0
e: 0
int(): 0
```

if, if...else and Nested if...else

```
// Program to print positive number entered by the user
// If the user enters a negative number, it is skipped
```

```
#include <iostream>
```

```
using std::cout;
```

```
using std::cin;
```

```
using std::endl;
```

```
int main()
```

```
{
```

```
    int number;
```

```
    cout << "Enter an integer: ";
```

```
    cin >> number;
```

```
    // checks if the number is positive
```

```
    if( number > 0 )
```

```
        cout << "You entered a positive integer: " << number << endl;
```

```
    cout << "This statement is always executed.\n";
```

```
}
```

```
Enter an integer: 23  
You entered a positive integer: 23  
This statement is always executed.
```

```
Enter an integer: -7  
This statement is always executed.
```

```
// Program to check whether an integer is nonnegative or negative
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int number;
```

```
    cout << "Enter an integer: ";
```

```
    cin >> number;
```

```
    if( number >= 0 )
```

```
        cout << "You entered a nonnegative integer: " << number << endl;
```

```
    else
```

```
        cout << "You entered a negative integer: " << number << endl;
```

```
    cout << "This line is always printed.\n";
```

```
}
```

```
Enter an integer: 23  
You entered a positive integer: 23  
This line is always printed.
```

```
Enter an integer: -7  
You entered a negative integer: -7  
This line is always printed.
```



```
// Program to check whether an integer is positive, negative or zero
```

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int number;
```

```
    cout << "Enter an integer: ";
```

```
    cin >> number;
```

```
    if( number > 0 )
```

```
        cout << "You entered a positive integer: " << number << endl;
```

```
    else if( number < 0 )
```

```
        cout << "You entered a negative integer: " << number << endl;
```

```
    else
```

```
        cout << "You entered 0." << endl;
```

```
    cout << "This line is always printed.\n";
```

```
}
```

```
Enter an integer: 23  
You entered a positive integer: 23  
This line is always printed.
```

```
Enter an integer: -7  
You entered a negative integer: -7  
This line is always printed.
```

```
Enter an integer: 0  
You entered 0.  
This line is always printed.
```

// Program to check whether an integer is positive, negative or zero

```
#include <iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter an integer: ";
    cin >> num;
    if( num != 0 ) // outer if condition
        if( num > 0 ) // inner if condition
            cout << "The number is positive.\n";
        else // inner else condition
            cout << "The number is negative.\n";
    else // outer else condition
        cout << "The number is 0.\n";

    cout << "This line is always printed.\n";
}
```

# Nested if . . . else Statements

```
if( a > 8 )  
    if( b > 8 )  
        cout << "a and b are > 8";  
else  
    cout << "a is <= 8";
```

The compiler interprets the above statement as

# Nested if . . . else Statements

```
if( a > 8 )  
    if( b > 8 )  
        cout << "a and b are > 8";  
else  
    cout << "a is <= 8";
```

The compiler interprets the above statement as

```
if( a > 8 )  
    if( b > 8 )  
        cout << "a and b are > 8";  
else  
    cout << "a is <= 8";
```

# Nested if . . . else Statements

- To force the nested if...else statement to execute as intended, use:

```
if( a > 8 )
{
    if( b > 8 )
        cout << "a and b are > 8";
}
else
    cout << "a is <= 8";
```

- Braces ({}) indicate that the second if statement is in the body of the first if and that the else is associated with the first if.

# Compound statements (Blocks)

- An example

```
if( grade >= 60 )  
    cout << "Passed.\n";  
else  
{  
    cout << "Failed.\n";  
    cout << "You must take this course again.\n";  
}
```

- Without the braces,

```
    cout << "You must take this course again.\n";  
always executed.
```

# Compound statements (Blocks)

- The `if` selection statement expects only one statement in its body.
- Similarly, the `if` and `else` parts of an `if...else` statement each expect only one body statement.
- To include several statements in the body of an `if` or in either part of an `if...else`, enclose the statements in braces (`{` and `}`).
- A set of statements contained within a pair of braces is called a **compound statement** or a **block**.



# C++ keywords

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>alignas</u> (C++11)<br><u>alignof</u> (C++11)<br><u>and</u><br><u>and_eq</u><br><u>asm</u><br><u>auto</u> (1)<br><u>bitand</u><br><u>bitor</u><br><u>bool</u><br><u>break</u><br><u>case</u><br><u>catch</u><br><u>char</u><br><u>char8_t</u> (C++20)<br><u>char16_t</u> (C++11)<br><u>char32_t</u> (C++11)<br><u>class</u> (1)<br><u>compl</u><br><u>concept</u> (C++20)<br><u>const</u><br><u>constexpr</u> (C++11)<br><u>constinit</u> (C++20)<br><u>const_cast</u><br><u>continue</u> | <u>co_await</u> (C++20)<br><u>co_return</u> (C++20)<br><u>co_yield</u> (C++20)<br><u>decltype</u> (C++11)<br><u>default</u> (1)<br><u>delete</u> (1)<br><u>do</u><br><u>double</u><br><u>dynamic_cast</u><br><u>else</u><br><u>enum</u><br><u>explicit</u><br><u>export</u> (1) (3)<br><u>extern</u> (1)<br><u>false</u><br><u>float</u><br><u>for</u><br><u>friend</u><br><u>goto</u><br><u>if</u><br><u>inline</u> (1)<br><u>int</u><br><u>long</u><br><u>mutable</u> (1) | <u>namespace</u><br><u>new</u><br><u>noexcept</u> (C++11)<br><u>not</u><br><u>not_eq</u><br><u>nullptr</u> (C++11)<br><u>operator</u><br><u>or</u><br><u>or_eq</u><br><u>private</u><br><u>protected</u><br><u>public</u><br><u>register</u> (2)<br><u>reinterpret_cast</u><br><u>requires</u> (C++20)<br><u>return</u><br><u>short</u><br><u>signed</u><br><u>sizeof</u> (1)<br><u>static</u><br><u>static_assert</u> (C++11)<br><u>static_cast</u><br><u>struct</u> (1)<br><u>switch</u> | <u>template</u><br><u>this</u> (4)<br><u>thread_local</u> (C++11)<br><u>throw</u><br><u>true</u><br><u>try</u><br><u>typedef</u><br><u>typeid</u><br><u>typename</u><br><u>union</u><br><u>unsigned</u><br><u>using</u> (1)<br><u>virtual</u><br><u>void</u><br><u>volatile</u><br><u>wchar_t</u><br><u>while</u><br><u>xor</u><br><u>xor_eq</u> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Conditional operator (?:)

```
number >= 0 ? cout << "positive" : cout << "negative";
```

```
cout << ( number >= 0 ? "positive" : "negative" );
```



condition



value if true



value if false

# Operators Precedence & Associativity Table

# Operators Precedence & Associativity

| Operator                                     | Associativity |
|----------------------------------------------|---------------|
| !<br>++ -- (post)<br>++ -- (pre) + - (unary) | Right to left |
| * / %                                        | Left to right |
| + - (binary)                                 | Left to right |
| < <= > >=                                    | Left to right |
| == !=                                        | Left to right |
| &&                                           | Left to right |
|                                              | Left to right |
| ? :                                          | Right to left |
| = *= /= %= += -=                             | Right to left |

# Assignment Operators

|                         |                |                      |
|-------------------------|----------------|----------------------|
| <code>a = a + 7;</code> | abbreviated to | <code>a += 7;</code> |
| <code>b = b - 4;</code> | abbreviated to | <code>b -= 4;</code> |
| <code>c = c * 5;</code> | abbreviated to | <code>c *= 5;</code> |
| <code>d = d / 3;</code> | abbreviated to | <code>d /= 3;</code> |
| <code>e = e % 9;</code> | abbreviated to | <code>e %= 9;</code> |

# Increment and Decrement Operators

- Increment operator (`++`)
  - Increment a variable by one
  - `i++`
    - Same as `i += 1`
- Decrement operator (`--`)
  - Decrement a variable by one
  - `i--`
    - Same as `i -= 1`

# Increment and Decrement Operators

- Preincrement
  - Variable changed before used in expression
    - Operator before variable (`++i` or `--i`)
- Postincrement
  - Incremented changed after expression
    - Operator after variable (`i++`, `i--`)

```
int main()
{
    int number;

    number = 7;
    cout << number << endl;
    cout << number++ << endl;
    cout << number << endl;

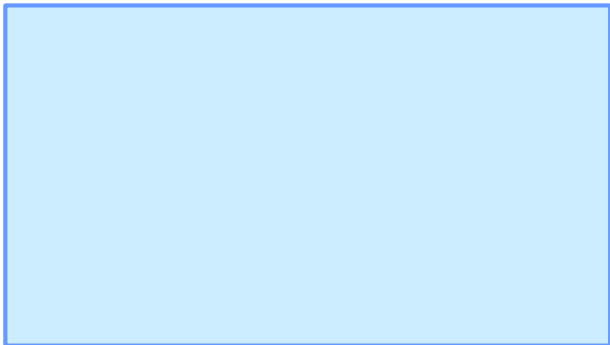
    cout << endl;

    number = 7;
    cout << number << endl;
    cout << ++number << endl;
    cout << number << endl;
}
```

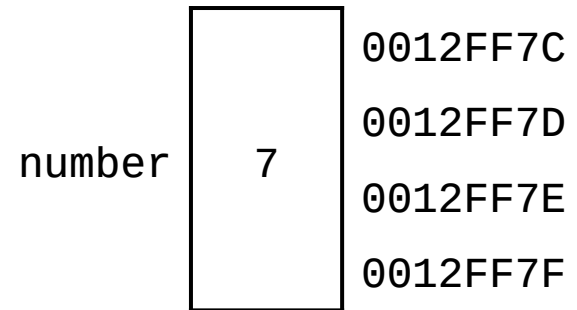


```
number = 7;  
cout << number << endl;  
cout << number++ << endl;  
cout << number << endl;
```

Output



Memory

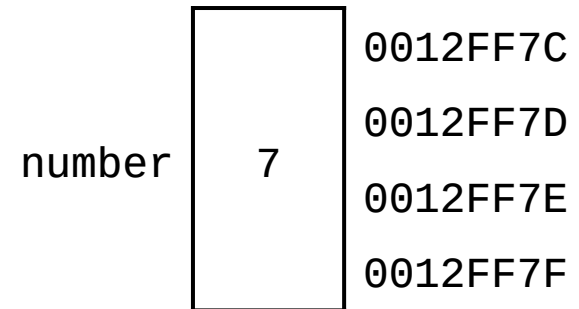


```
number = 7;  
cout << number << endl;  
cout << number++ << endl;  
cout << number << endl;
```

## Output

7

## Memory

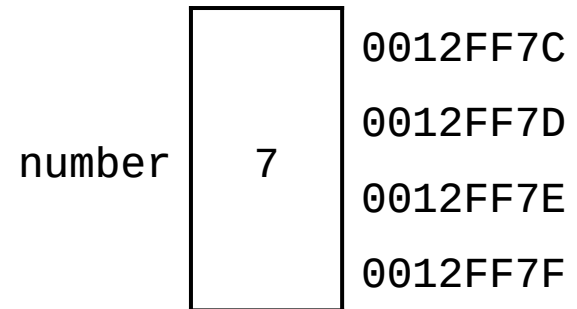


```
number = 7;  
cout << number << endl;  
cout << number++ << endl;  
cout << number << endl;
```

## Output

```
7  
7
```

## Memory

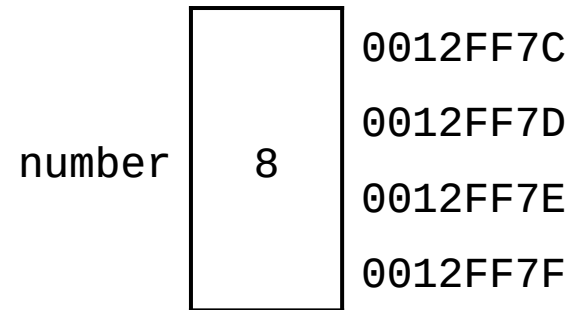


```
number = 7;  
cout << number << endl;  
cout << number++ << endl;  
cout << number << endl;
```

## Output

```
7  
7
```

## Memory

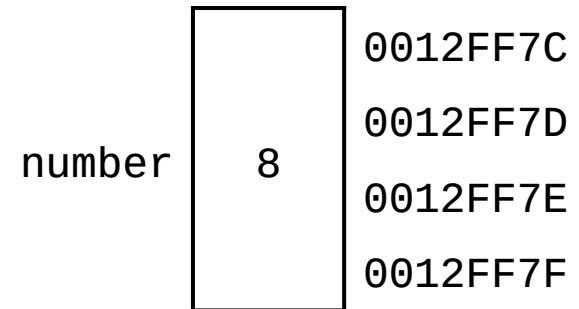


```
number = 7;  
cout << number << endl;  
cout << number++ << endl;  
cout << number << endl;
```

## Output

```
7  
7  
8
```

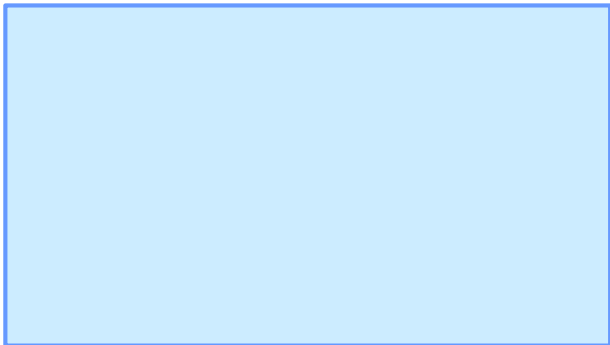
## Memory



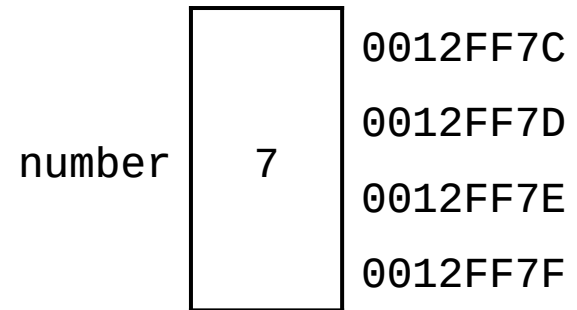


```
number = 7;  
cout << number << endl;  
cout << ++number << endl;  
cout << number << endl;
```

Output



Memory

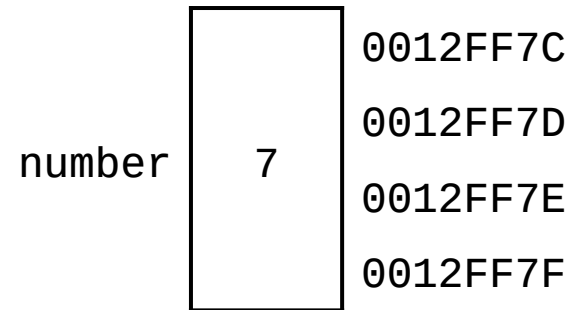


```
number = 7;  
cout << number << endl;  
cout << ++number << endl;  
cout << number << endl;
```

## Output

7

## Memory



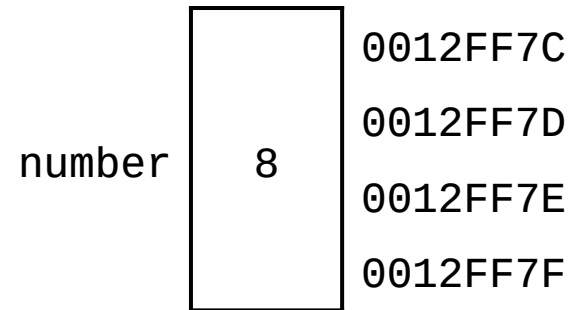


```
number = 7;  
cout << number << endl;  
cout << ++number << endl;  
cout << number << endl;
```

## Output

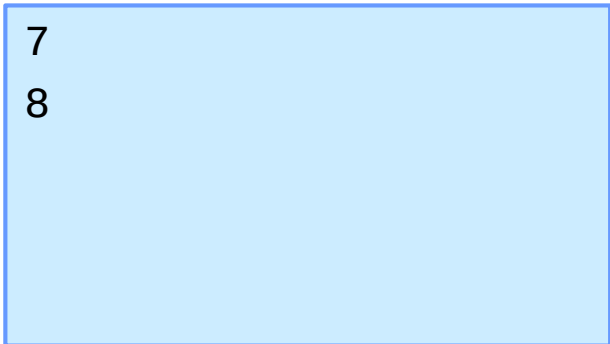
7

## Memory



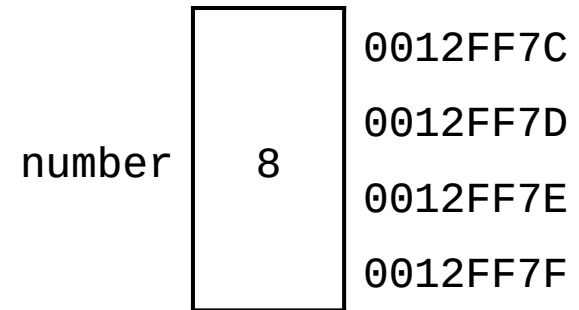
```
number = 7;  
cout << number << endl;  
cout << ++number << endl;  
cout << number << endl;
```

## Output



7  
8

## Memory

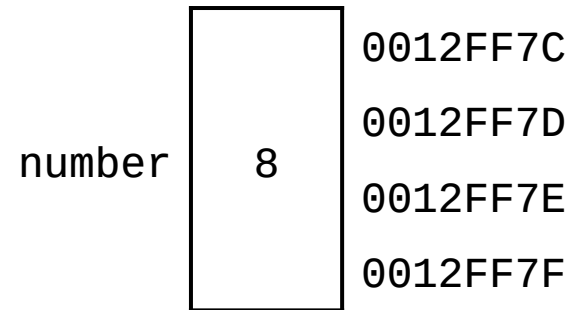


```
number = 7;  
cout << number << endl;  
cout << ++number << endl;  
cout << number << endl;
```

## Output

```
7  
8  
8
```

## Memory



# Increment and Decrement Operators

- When you increment (++) or decrement (- -) a variable in a statement by itself, the preincrement and postincrement forms have the same effect

```
++number;  
cout << number;
```

and

```
number++;  
cout << number;
```

print the same value.

# Operators Precedence & Associativity

| Operator                                     | Associativity |
|----------------------------------------------|---------------|
| !<br>++ -- (post)<br>++ -- (pre) + - (unary) | Right to left |
| * / %                                        | Left to right |
| + - (binary)                                 | Left to right |
| << >>                                        | Left to right |
| < <= > >=                                    | Left to right |
| == !=                                        | Left to right |
| &&                                           | Left to right |
|                                              | Left to right |
| ? :                                          | Right to left |
| = *= /= %= += -=                             | Right to left |

| Data types                                                         |                                                              |
|--------------------------------------------------------------------|--------------------------------------------------------------|
| <code>long double (8 bytes)</code>                                 | 絕對值範圍大約是 $2.2 \cdot 10^{-308} \sim 1.8 \cdot 10^{308}$       |
| <code>double (8 bytes)</code>                                      | 絕對值範圍大約是 $2.2 \cdot 10^{-308} \sim 1.8 \cdot 10^{308}$       |
| <code>float (4 bytes)</code>                                       | 絕對值範圍大約是 $1.2 \cdot 10^{-38} \sim 3.4 \cdot 10^{38}$         |
| <code>unsigned long long int (unsigned long long) (8 bytes)</code> | $0 \sim 2^{64}-1$                                            |
| <code>long long int (long long) (8 bytes)</code>                   | $-2^{63} \sim 2^{63}-1$                                      |
| <code>unsigned long int (unsigned long) (4 bytes)</code>           | $0 \sim 2^{32}-1$                                            |
| <code>long int (long) (4 bytes)</code>                             | $-2^{31} \sim 2^{31}-1$ (2147483647)                         |
| <code>unsigned int (unsigned) (4 bytes)</code>                     | $0 \sim 2^{32}-1$ (4294967295)                               |
| <code>int (4 bytes)</code>                                         | $-2^{31} \sim 2^{31}-1$ (2147483647)                         |
| <code>unsigned short int (unsigned short) (2)</code>               | $0 \sim 2^{16}-1$ (65535)                                    |
| <code>short int (short) (2 bytes)</code>                           | $-2^{15} \sim 2^{15}-1$ (32767)                              |
| <code>unsigned char (1 byte)</code>                                | $0 \sim 2^8-1$ (0 ~ 255)                                     |
| <code>char (1 byte)</code>                                         | $-2^7 \sim 2^7-1$ (-128 ~ 127)                               |
| <code>bool (1 byte)</code>                                         | ( <code>false</code> becomes 0, <code>true</code> becomes 1) |

| binary representation               | int         |               |
|-------------------------------------|-------------|---------------|
| 01111111 11111111 11111111 11111111 | 2147483647  | $2^{31} - 1$  |
| 01111111 11111111 11111111 11111110 | 2147483646  | $2^{31} - 2$  |
| 01111111 11111111 11111111 11111100 | 2147483645  | $2^{31} - 3$  |
| .....                               | .....       |               |
| 00000000 00000000 00000000 00000011 | 3           |               |
| 00000000 00000000 00000000 00000010 | 2           |               |
| 00000000 00000000 00000000 00000001 | 1           |               |
| 00000000 00000000 00000000 00000000 | 0           |               |
| 11111111 11111111 11111111 11111111 | -1          |               |
| 11111111 11111111 11111111 11111110 | -2          |               |
| 11111111 11111111 11111111 11111101 | -3          |               |
| .....                               | .....       |               |
| 10000000 00000000 00000000 00000010 | -2147483646 | $-2^{31} + 2$ |
| 10000000 00000000 00000000 00000001 | -2147483647 | $-2^{31} + 1$ |
| 10000000 00000000 00000000 00000000 | -2147483648 | $-2^{31}$     |

| binary representation               | unsigned int |              |
|-------------------------------------|--------------|--------------|
| 11111111 11111111 11111111 11111111 | 4294967295   | $2^{32} - 1$ |
| 11111111 11111111 11111111 11111110 | 4294967294   | $2^{32} - 2$ |
| 11111111 11111111 11111111 11111101 | 4294967293   | $2^{32} - 3$ |
| .....                               | .....        |              |
| 10000000 00000000 00000000 00000010 | 2147483650   | $2^{31} + 2$ |
| 10000000 00000000 00000000 00000001 | 2147483649   | $2^{31} + 1$ |
| 10000000 00000000 00000000 00000000 | 2147483648   | $2^{31}$     |
| 01111111 11111111 11111111 11111111 | 2147483647   | $2^{31} - 1$ |
| 01111111 11111111 11111111 11111110 | 2147483646   | $2^{31} - 2$ |
| 01111111 11111111 11111111 11111100 | 2147483645   | $2^{31} - 3$ |
| .....                               | .....        |              |
| 00000000 00000000 00000000 00000011 | 3            |              |
| 00000000 00000000 00000000 00000010 | 2            |              |
| 00000000 00000000 00000000 00000001 | 1            |              |
| 00000000 00000000 00000000 00000000 | 0            |              |



| binary representation               | int         | unsigned int |
|-------------------------------------|-------------|--------------|
| 01111111 11111111 11111111 11111111 | 2147483647  | 2147483647   |
| 01111111 11111111 11111111 11111110 | 2147483646  | 2147483646   |
| 01111111 11111111 11111111 11111100 | 2147483645  | 2147483645   |
| .....                               | .....       | .....        |
| 00000000 00000000 00000000 00000011 | 3           | 3            |
| 00000000 00000000 00000000 00000010 | 2           | 2            |
| 00000000 00000000 00000000 00000001 | 1           | 1            |
| 00000000 00000000 00000000 00000000 | 0           | 0            |
| 11111111 11111111 11111111 11111111 | -1          | 4294967295   |
| 11111111 11111111 11111111 11111110 | -2          | 4294967294   |
| 11111111 11111111 11111111 11111101 | -3          | 4294967293   |
| .....                               | .....       | .....        |
| 10000000 00000000 00000000 00000010 | -2147483646 | 2147483650   |
| 10000000 00000000 00000000 00000001 | -2147483647 | 2147483649   |
| 10000000 00000000 00000000 00000000 | -2147483648 | 2147483648   |

| hexadecimal        | long long int        | unsigned long long   |
|--------------------|----------------------|----------------------|
| 7FFFFFFFFFFFFFFFFF | 9223372036854775807  | 9223372036854775807  |
| 7FFFFFFFFFFFFFFFFE | 9223372036854775806  | 9223372036854775806  |
| 7FFFFFFFFFFFFFFFFD | 9223372036854775805  | 9223372036854775805  |
| .....              | .....                | .....                |
| 00000000000000003  | 3                    | 3                    |
| 00000000000000002  | 2                    | 2                    |
| 00000000000000001  | 1                    | 1                    |
| 00000000000000000  | 0                    | 0                    |
| FFFFFFFFFFFFFFFFF  | -1                   | 18446744073709551615 |
| FFFFFFFFFFFFFFFFE  | -2                   | 18446744073709551614 |
| FFFFFFFFFFFFFFFFD  | -3                   | 18446744073709551613 |
| .....              | .....                | .....                |
| 80000000000000002  | -9223372036854775806 | 9223372036854775810  |
| 80000000000000001  | -9223372036854775807 | 9223372036854775809  |
| 80000000000000000  | -9223372036854775808 | 9223372036854775808  |

# Maximum Finding

```
int main()  
{  
    int number1;  
    int number2;  
  
    cin >> number1;  
    cin >> number2;  
  
    if(                )  
        ;  
    else                ;  
}
```

# Maximum Finding

```
int main()
{
    int number1;
    int number2;

    cin >> number1;
    cin >> number2;

    if( number1 >= number2 )
        ;
    else
        ;
}
```

# Maximum Finding

```
int main()  
{  
    int number1;  
    int number2;  
  
    cin >> number1;  
    cin >> number2;  
  
    if( number1 >= number2 )  
        cout << number1;  
    else  
        ;  
}
```

# Maximum Finding

```
int main()
{
    int number1;
    int number2;

    cin >> number1;
    cin >> number2;

    if( number1 >= number2 )
        cout << number1;
    else
        cout << number2;
}
```

# Maximum Finding

```
int main()
{
    int number1;
    int number2;
    int number3;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    if(
        if(
            ;
        else
            ;
    else
}
}
```

# Maximum Finding

```
int main()
{
    int number1;
    int number2;
    int number3;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    if( number1 >= number2 )
        if(
            ;
        else
            ;
    else
        ;
}
```



# Maximum Finding

```
int main()
{
    int number1;
    int number2;
    int number3;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    if( number1 >= number2 )
        if( number1 >= number3 )
            ;
        else
            ;
    else
        ;
}
```

# Maximum Finding

```
int main()
{
    int number1;
    int number2;
    int number3;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    if( number1 >= number2 )
        if( number1 >= number3 )
            cout << number1;
        else
            ;
    else

```

}

# Maximum Finding

```
int main()
{
    int number1;
    int number2;
    int number3;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    if( number1 >= number2 )
        if( number1 >= number3 )
            cout << number1;
        else
            cout << number3;
    else

```

}

# Maximum Finding

```
int main()
{
    int number1;
    int number2;
    int number3;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    if( number1 >= number2 )
        if( number1 >= number3 )
            cout << number1;
        else
            cout << number3;
    else
        if( number2 >= number3 )
            cout << number2;
        else
            cout << number3;
}
```

```

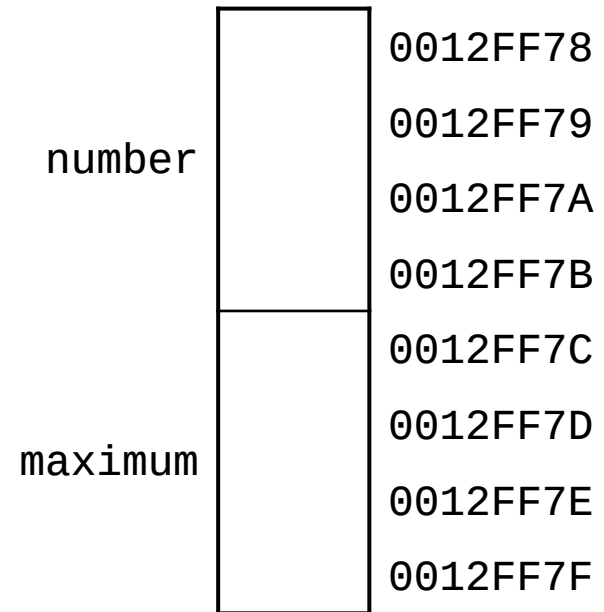
int main()
{
    int number1;
    int number2;
    int number3;
    int number4;
    cin >> number1;
    cin >> number2;
    cin >> number3;
    cin >> number4;

    if( number1 >= number2 )
        if( number1 >= number3 )
            if( number1 >= number4 )
                cout << number1;
            else
                cout << number4;
        else
            if( number3 >= number4 )
                cout << number3;
            else
                cout << number4;
    else
        if( number2 >= number3 )
            if( number2 >= number4 )
                cout << number2;
            else
                cout << number4;
        else
            if( number3 >= number4 )
                cout << number3;
            else
                cout << number4;
}

```

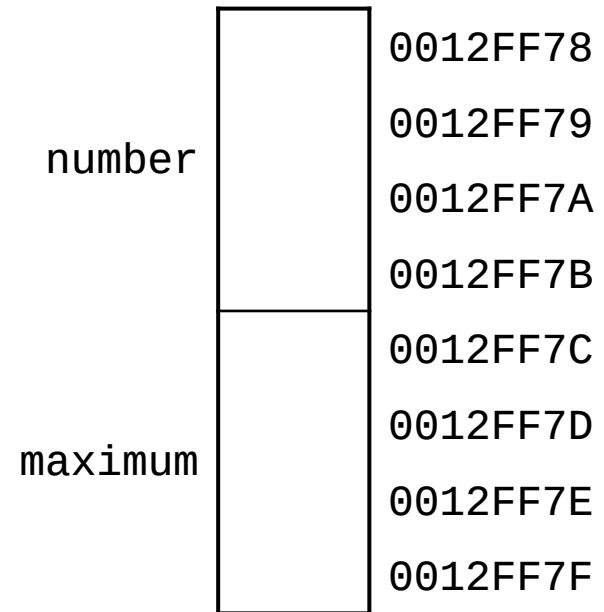
# Maximum Finding

```
cin >> maximum;  
cin >> number;
```



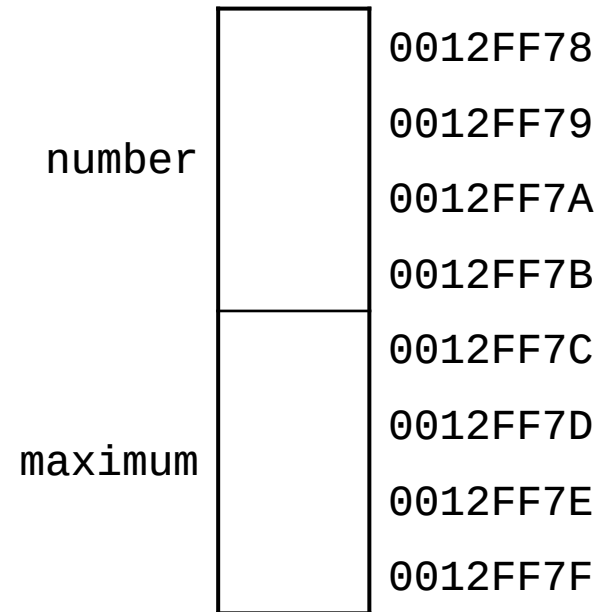
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )
```



# Maximum Finding

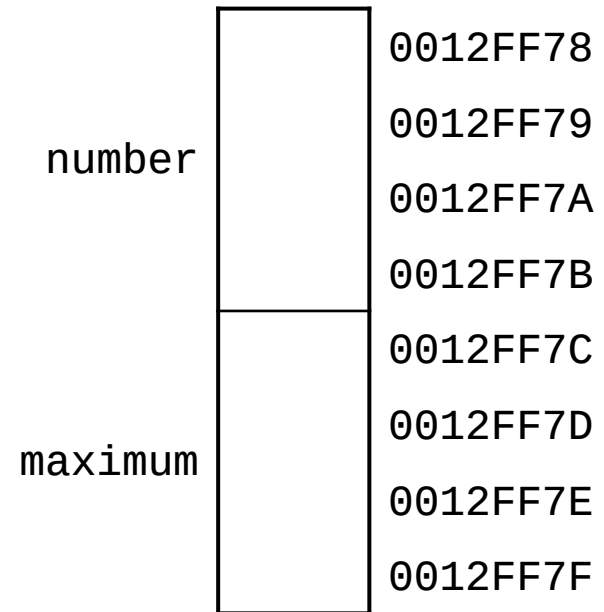
```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;
```





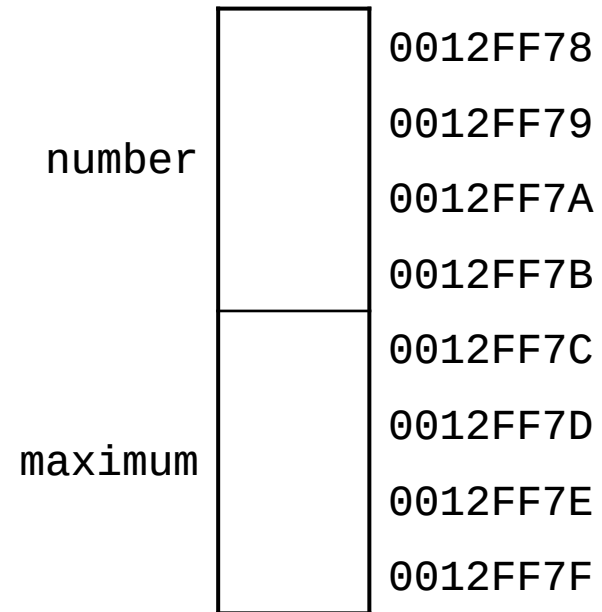
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )
```



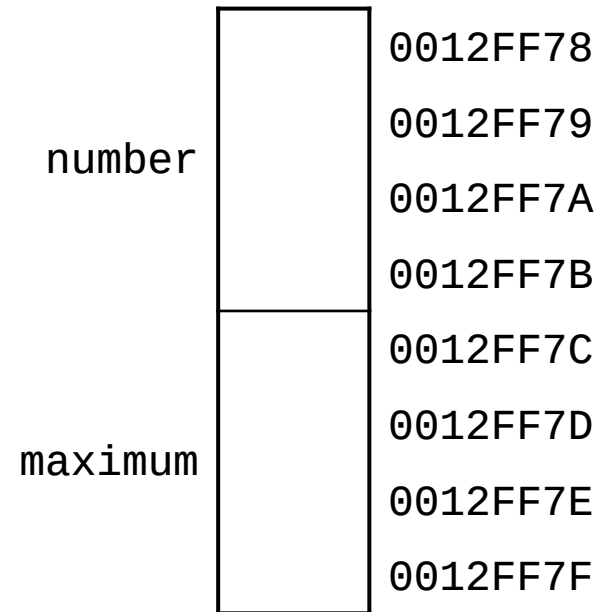
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;
```



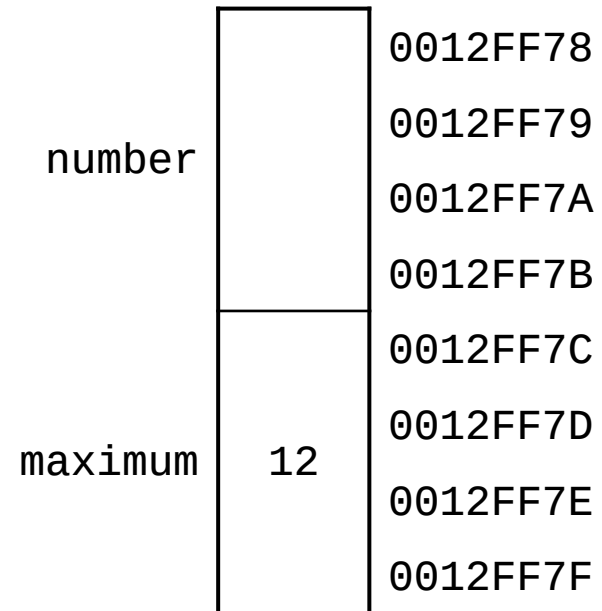
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```



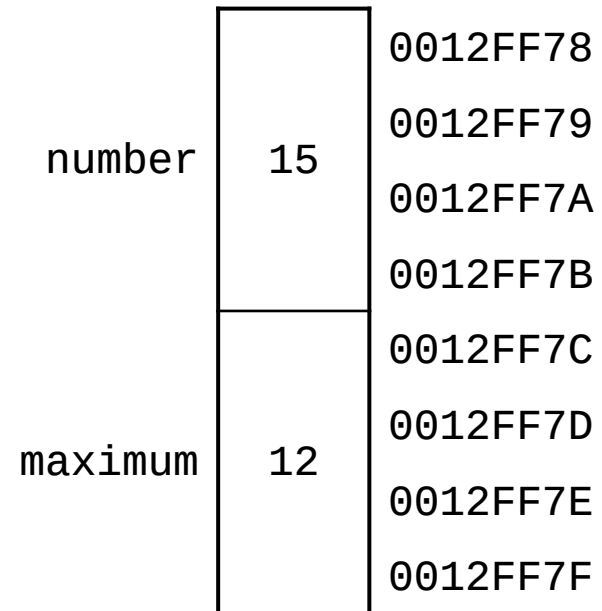
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```



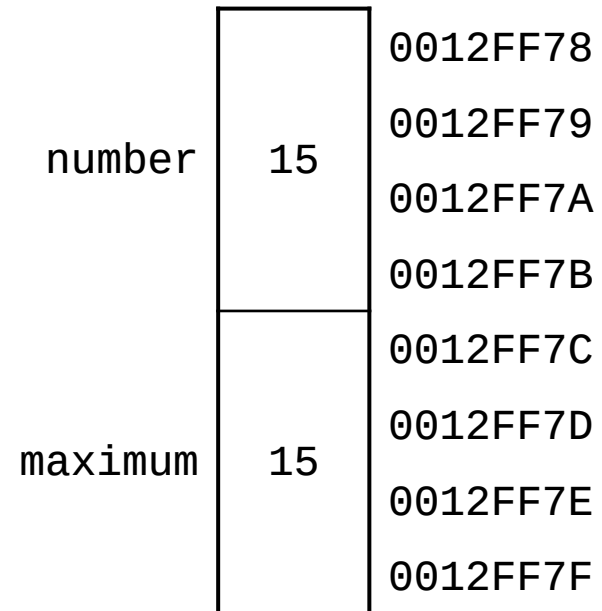
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```



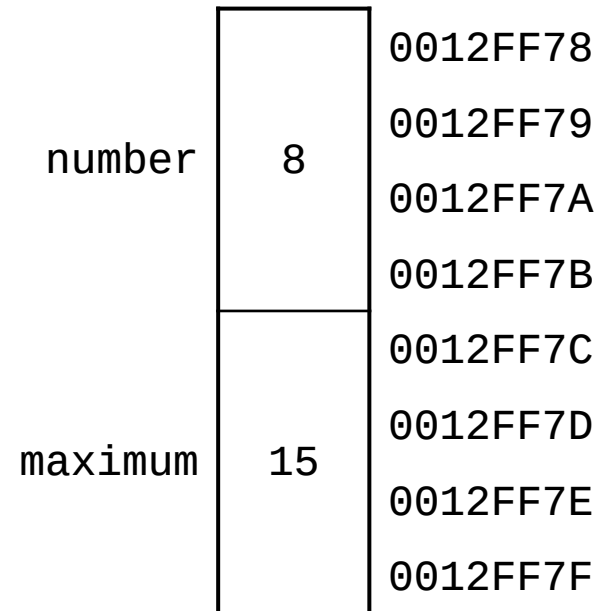
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```



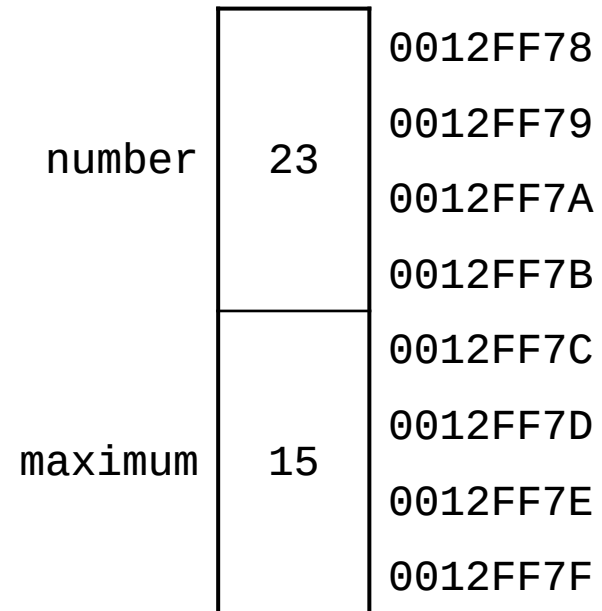
# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```



# Maximum Finding

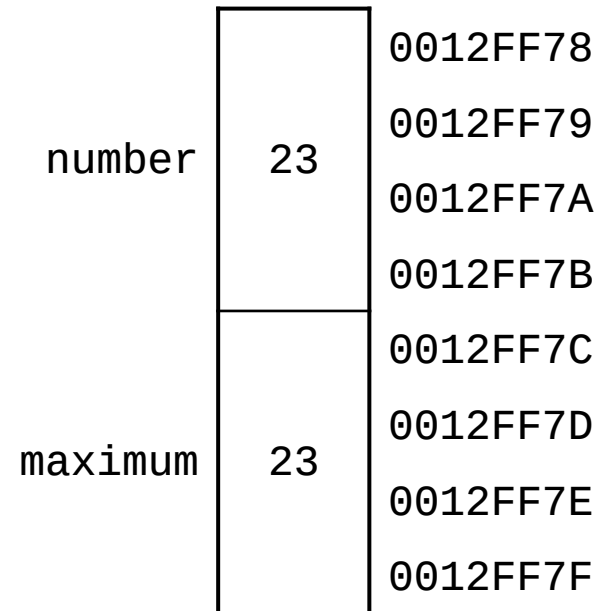
```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```





# Maximum Finding

```
cin >> maximum;  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;  
  
cin >> number;  
if( number > maximum )  
    maximum = number;
```



# Introduction to Functions

```
int main()
{
    int number1;
    int number2;
    int number3;

    cout << "Enter three numbers: ";
    cin >> number1 >> number2 >> number3;

    int max = number1; // assume number1 is largest

    if( number2 > max ) // if number2 is larger,
        max = number2; // assign number2 to max

    if( number3 > max ) // if number3 is larger,
        max = number3; // assign number3 to max

    cout << "Maximum is: " << max << endl;
}
```

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum();
}
```

```
void maximum()
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    cout << max << endl;
}
```

max 



 0012FF10

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF78 |
| number2 | 9 | 0012FF80 |
| number1 | 4 | 0012FF88 |

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    cout << max << endl;
}
```

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum( number1, number2, number3 );
}
```

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF78 |
| number2 | 9 | 0012FF80 |
| number1 | 4 | 0012FF88 |

```
void maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    cout << max << endl;
}
```

|         |   |          |
|---------|---|----------|
| max     |   | 0012FF10 |
| number1 | 4 | 0012FF14 |
| number2 | 9 | 0012FF18 |
| number3 | 6 | 0012FF1C |

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    cout << max << endl;
}
```

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

```

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF78 |
| number2 | 9 | 0012FF80 |
| number1 | 4 | 0012FF88 |

```

int maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    return max;
}

```

|         |   |          |
|---------|---|----------|
| max     |   | 0012FF10 |
| number1 | 4 | 0012FF14 |
| number2 | 9 | 0012FF18 |
| number3 | 6 | 0012FF1C |

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    cout << max << endl;
}

```

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

```

```

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}

```

|     |   |          |
|-----|---|----------|
| max |   | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF78 |
| number2 | 9 | 0012FF80 |
| number1 | 4 | 0012FF88 |

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    cout << max << endl;
}

```

```

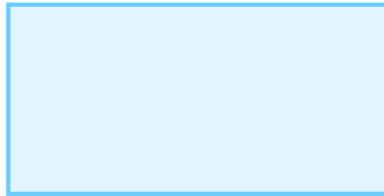
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max;
}

```

```

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}

```

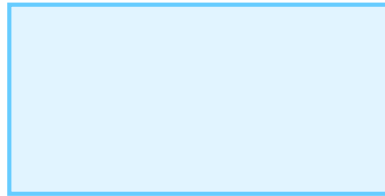


Output

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max;
}

```



Output

```

                                  
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}

```



```

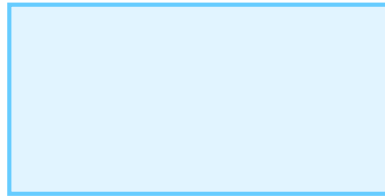
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max;
}

```

```

                                4    9    6
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
                                9

```



Output

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max;
}

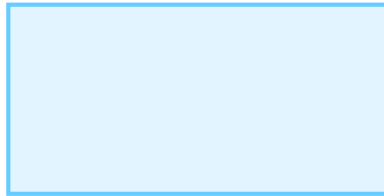
```

9

4

9

6



Output

```

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}

```

```

int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max;
}

```

9

4

9

6

9

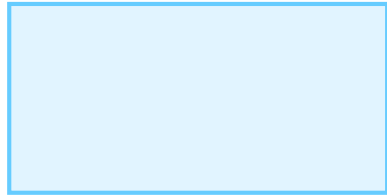
Output

```

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}

```

# Output

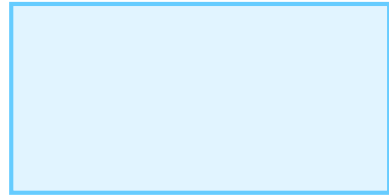


```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}
```

```
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

# Output



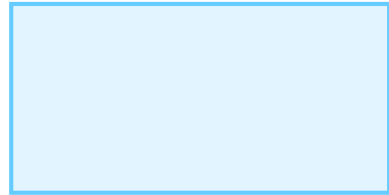
```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}
```

```
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|     |  |          |
|-----|--|----------|
| max |  | 0012FF10 |
| a   |  | 0012FF14 |
| b   |  | 0012FF18 |
| c   |  | 0012FF1C |

# Output



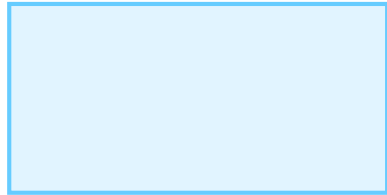
```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}
```

```
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |

# Output



```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}
```

```
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register | 9 |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |

# Output

9

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}
```

```
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

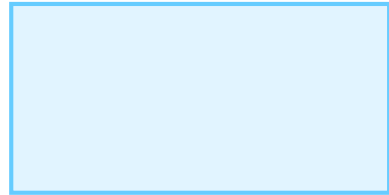
|          |   |          |
|----------|---|----------|
| register | 9 |          |
| max      | 9 | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |





# Output

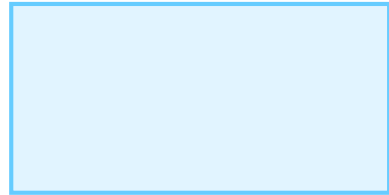


```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

int maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

# Output



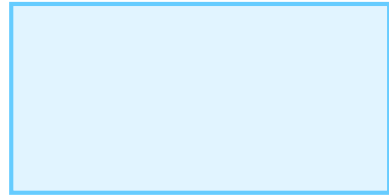
```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

int maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|         |  |          |
|---------|--|----------|
| max     |  | 0012FF10 |
| number1 |  | 0012FF14 |
| number2 |  | 0012FF18 |
| number3 |  | 0012FF1C |

# Output



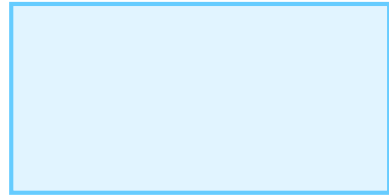
```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

int maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|         |   |          |
|---------|---|----------|
| max     | 9 | 0012FF10 |
| number1 | 4 | 0012FF14 |
| number2 | 9 | 0012FF18 |
| number3 | 6 | 0012FF1C |

# Output



```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

int maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register | 9 |          |
| max      |   | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|         |   |          |
|---------|---|----------|
| max     | 9 | 0012FF10 |
| number1 | 4 | 0012FF14 |
| number2 | 9 | 0012FF18 |
| number3 | 6 | 0012FF1C |

# Output

9

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    int max = maximum( number1, number2, number3 );
    cout << max << endl;
}

int maximum( int number1, int number2, int number3 )
{
    int max = number1;
    if( number2 > max )
        max = number2;
    if( number3 > max )
        max = number3;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register | 9 |          |
| max      | 9 | 0012FF70 |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|         |   |          |
|---------|---|----------|
| max     | 9 | 0012FF10 |
| number1 | 4 | 0012FF14 |
| number2 | 9 | 0012FF18 |
| number3 | 6 | 0012FF1C |

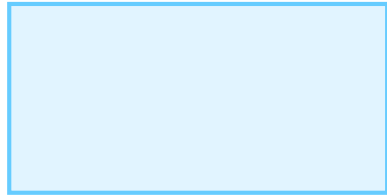


```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    cout << maximum( number1, number2, number3 ) << endl;
}

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```



# Output

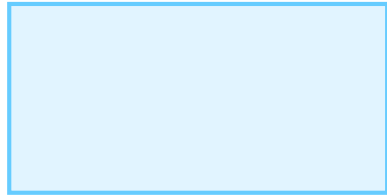


|          |   |          |
|----------|---|----------|
| register |   |          |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    cout << maximum( number1, number2, number3 ) << endl;
}

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

# Output



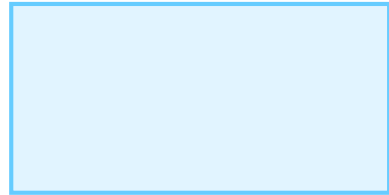
```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    cout << maximum( number1, number2, number3 ) << endl;
}

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|     |  |          |
|-----|--|----------|
| max |  | 0012FF10 |
| a   |  | 0012FF14 |
| b   |  | 0012FF18 |
| c   |  | 0012FF1C |

# Output



```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    cout << maximum( number1, number2, number3 ) << endl;
}

int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

|          |   |          |
|----------|---|----------|
| register |   |          |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |

# Output

9

|          |   |          |
|----------|---|----------|
| register | 9 |          |
| number3  | 6 | 0012FF74 |
| number2  | 9 | 0012FF78 |
| number1  | 4 | 0012FF7C |

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    cout << maximum( number1, number2, number3 ) << endl;
}
```

```
int maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    return max;
}
```

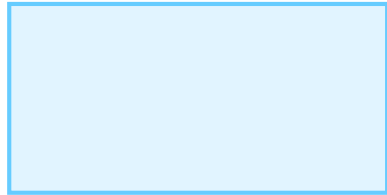
|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |



```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum( number1, number2, number3 );
}

void maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    cout << max << endl;
}
```

# Output

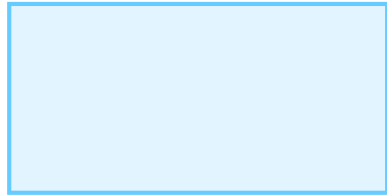


|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF74 |
| number2 | 9 | 0012FF78 |
| number1 | 4 | 0012FF7C |

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum( number1, number2, number3 );
}

void maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    cout << max << endl;
}
```

# Output



```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum( number1, number2, number3 );
}
```

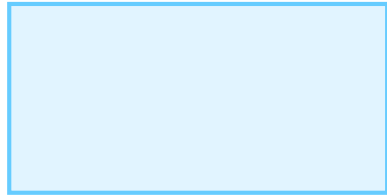
```
void maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    cout << max << endl;
}
```

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF74 |
| number2 | 9 | 0012FF78 |
| number1 | 4 | 0012FF7C |

|     |  |          |
|-----|--|----------|
| max |  | 0012FF10 |
| a   |  | 0012FF14 |
| b   |  | 0012FF18 |
| c   |  | 0012FF1C |



# Output



```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum( number1, number2, number3 );
}
```

```
void maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    cout << max << endl;
}
```

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF74 |
| number2 | 9 | 0012FF78 |
| number1 | 4 | 0012FF7C |

|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |

# Output

9

```
int main()
{
    int number1, number2, number3;
    cin >> number1 >> number2 >> number3;
    maximum( number1, number2, number3 );
}
```

```
void maximum( int a, int b, int c )
{
    int max = a;
    if( b > max )
        max = b;
    if( c > max )
        max = c;
    cout << max << endl;
}
```

|         |   |          |
|---------|---|----------|
| number3 | 6 | 0012FF74 |
| number2 | 9 | 0012FF78 |
| number1 | 4 | 0012FF7C |

|     |   |          |
|-----|---|----------|
| max | 9 | 0012FF10 |
| a   | 4 | 0012FF14 |
| b   | 9 | 0012FF18 |
| c   | 6 | 0012FF1C |